RESEARCH MANUSCRIPT

Asset Price Bubbles

Theory, Signals, and Traded Contracts

Rebalancing and identification
Daily and intraday monitoring
Leveraged funds, options, and constituents

A unified account of the research, with economic explanations, principal mathematical results, and evidence from simulations and market data.

Draft prepared 9 October 2026

Evidence through the research snapshot documented in Appendix C.

Abstract

In the strict-local-martingale theory of asset prices, a bubble is a shortfall of expected future value under a pricing measure relative to the current price. This manuscript studies how that valuation concept relates to patterns in observed prices and to the wealth produced by trading strategies. We classify constant-proportion wealth by exposure and the asset's behavior near zero and at very high prices. Continuous rebalancing can create a strict local martingale from a true-martingale asset, and the classification can reverse when rebalancing occurs on a finite grid. We also establish limits on identification from price paths and examine what liquidation rules, option contracts, and constituent information add to the valuation problem.

The empirical work develops daily and intraday signals of return variance rising with the price level. This pattern has a connection to bubble mechanisms in both local-volatility and stochastic-volatility models. The measurement problem is to distinguish it from changes in background volatility. The daily method uses a predictable market volatility scale; the intraday method compares price movements within each session while updating its background state between sessions. We evaluate the methods in simulations and historical records, with particular attention to how calibration changes their alarm patterns.

The results show substantial sensitivity in the intended simulation families and distinct behavior on market data. In shared-market-scale and constant-elasticity simulations, the daily statistic detects 39–72% of the specified alternatives, compared with 7–14% for a calibrated explosive-price benchmark. It remains relatively quiet in crisis cohorts and flags all four studied homebuilders about 400 trading sessions before their 2005 peaks, versus one for the benchmark. The intraday score adds out-of-sample drawdown information to that benchmark. These findings support the signals as measurements of price–variance behavior, with the theory specifying the additional assumptions needed to interpret such behavior as a pricing bubble.

How to read the manuscript

The four parts answer different questions within one framework. Part I defines the economic object and establishes what rebalancing and observation can change. Part II introduces the proposed signals, their calibration, and their daily and intraday evidence. Part III follows the theory into finite trading and liquidation. Part IV asks what options and index constituents add to prices alone. A reader interested primarily in monitoring can begin with Section 1, then read Part II; the preceding theory explains the limits on interpretation.

Section 2 introduces the relevant literature and explains how the contributions build on it. The main text includes the short arguments that explain the results. Longer derivations, implementation specifications, and the research-to-manuscript map are in the appendices. Experiment numbers identify reproducible research records, not a sequence the reader must follow. No new empirical experiment is introduced in this manuscript: the tables and figures reorganize existing results. The selected generated tables are read directly from the stored experiment summaries at build time.

Contents

Abstract i

How to read the manuscript i

1 The questions and the contributions 1

1.1 What this research adds . . . 2

1.2 How to assess the evidence . . . 2

2 Background and relation to existing work 3

2.1 From fundamental value to a pricing defect . . . 3

2.2 What volatility can reveal about a bubble . . . 3

2.3 Explosive prices, episode dating, and crash risk . . . 4

2.4 Separating price dependence from background volatility . . . 5

2.5 Investment, rebalancing, and leveraged funds . . . 5

2.6 What options and constituents add . . . 6

I Theory and the Limits of Identification 7

3 Constant exposure and the two endpoints 7

3.1 Why one-half appears . . . 8

3.2 Long horizons, changing exposure, and independent assets . . . 8

4 What prices identify 9

4.1 Sin's model: feedback and the pricing measure . . . 10

4.2 What is positively identified inside the range . . . 10

4.3 Why sampling frequency is not a universal cure . . . 11

II Price-Level Volatility Signals 12

5 The signal and the background variance 12

5.1 One conditional-return model on both sides . . . 12

5.2 Daily market-scale construction . . . 13

5.3 Intraday session-anchor construction . . . 13

5.4 Why the earlier likelihood score was replaced . . . 14

6 Calibration is part of the method 14

6.1 The strict rank rule and its premise . . . 14

6.2 A measured calibration effect on GSADF . . . 15
6.3 Exact-model guarantees and empirical calibration . . . 15

7 Simulation: recovery, power, and confounding 16
7.1 The daily market-scale method in its intended family . . . 16
7.2 What a positive fitted exponent can mean . . . 17
7.3 Calibration transport can dominate the result . . . 17
7.4 What the earlier exact-model experiments establish . . . 17

8 Daily observations in the market 18
8.1 Data and comparison design . . . 18
8.2 The daily market-scale signal changes the alarm pattern . . . 19
8.3 Homebuilders: a coherent early sector signal . . . 19
8.4 How this differs from the repaired own-state score . . . 21
8.5 What the earlier alarms contain . . . 21
8.6 Implied volatility as a different background scale . . . 21

9 What five-minute data add 22
9.1 Within-session variance behavior . . . 22
9.2 A retrospective replay of the monitor . . . 22
9.3 Incremental out-of-sample drawdown information . . . 23

10 Reading and using an alarm 23

III Rebalancing and Traded Funds 25

11 Finite rebalancing changes the classification 25
11.1 Finite grids with exposure between zero and one . . . 25
11.2 Refinement and the size of the discrepancy . . . 26

12 Leverage, inverse exposure, and liquidation 27
12.1 Inverse funds under an explicit floor rule . . . 27
12.2 Leveraged-long funds . . . 27
12.3 Costs are not a bubble . . . 27

13 What discrete monitoring misses 28

IV Options and Constituent Information 29

14 Option prices do not have a unique bubble interpretation **29**
14.1 Exercise value and the parity residual 29

15 Which option observations are informative? **30**
15.1 The lowest strike matters 30
15.2 Do options on leveraged funds add information? 30
15.3 Continuous exercise creates another limiting distinction 31

16 From an index to its constituents **31**
16.1 Ordinary weights can hide all of the effect 32
16.2 A conditional recovery formula 32
16.3 What the synthetic estimation says 33
16.4 What a basket budget can and cannot identify 33

17 What the four parts establish **33**

A Notation and algorithm specifications **35**
A.1 Common notation 35
A.2 Student-t density and the numerical profile 35
A.3 Daily implementation 36
A.4 Intraday implementation 36
A.5 Comparators and differences between experiments 37

B Selected derivations **38**
B.1 The endpoint calculation 38
B.2 From tail completion to testing 38
B.3 An honest likelihood-ratio guarantee 38
B.4 The rank calibration bound 39

C Research provenance and verification scope **40**
C.1 Principal mathematical sources within the project 40
C.2 Empirical source map 41
C.3 What Lean verifies 42
C.4 Literature positioning 42

References **44**

1 The questions and the contributions

An investor confronted with a rapidly appreciating stock might ask whether the price is detached from fundamental value, whether volatility rises as the price rises, whether a crash is becoming more likely, and whether a leveraged position changes those risks. These questions concern different aspects of the same episode. Understanding their connections helps determine what to measure and how to interpret the result.

We focus on a particular pattern: return variance increasing with the price level. It appears in models where the price is a strict local martingale, and it offers something concrete to look for in market data. Measuring it requires separating movements associated with the stock's price from changes in market conditions and background volatility. The daily and intraday signals developed here make that separation in different ways. Alongside them, the theoretical analysis explains how the bubble concept changes when we move from an asset to a rebalanced portfolio or an option contract.

Our starting point is the strict-local-martingale definition of a bubble [14]. In discounted units, under a specified pricing measure Q, a nonnegative local martingale S is a supermartingale. Its finite-horizon defect is

$$b_S(T) = S_0 - \mathbb{E}_Q[S_T] \geq 0. \tag{1}$$

A positive defect means that S fails to be a true martingale on that horizon. The physical measure P describes the distribution from which historical observations are drawn. An expectation under Q is a valuation object, not an estimate of the investor's average realized return under P.

A widely used approach to monitoring bubbles looks for explosive price growth: intervals on which prices behave like an autoregression with a root greater than one. The generalized supremum augmented Dickey–Fuller test (GSADF) is a prominent example [21]. We use an empirically calibrated version as a benchmark. Its focus on price dynamics provides a useful comparison with our focus on the relationship between price and return variance. A third question, whether a signal predicts a future drawdown, concerns outcomes under P. The literature already documents that large run-ups and their characteristics can carry crash information without run-ups necessarily predicting low average returns [9]. Forecasting is therefore one way to examine usefulness, not the definition of the object studied here.

Object	Evidence it requires	What a positive finding means
Pricing defect	A pricing law and a valuation convention	A shortfall of terminal expectation relative to current value
Explosive behavior	A specified time-series null and calibration	Departure toward explosive price dynamics in that testing framework
Price-level variance dependence	A variance model, nuisance treatment, and calibration	An observed pattern better fitted by positive level dependence
Future loss risk	A forecast design and genuinely later outcomes	An association with losses over a specified horizon

Table 1: Four questions often grouped under the word "bubble." A result in one row does not automatically establish a result in another.

1.1 What this research adds

The theoretical centerpiece is a classification of constant-proportion wealth. An investor who continually restores a fixed fraction of wealth to an asset can inherit its strictness or acquire strictness through a different mechanism: rebalancing into an asset that can reach zero. The result depends on exposure and on the asset's behavior at both ends of its price range. Finite rebalancing and liquidation change the answer, which gives the classification a direct interpretation for traded funds. The result applies established martingale criteria; its prior-art comparison is incomplete, and we make no separate novelty claim for the finite-trading results.

The identification results explain what observed prices can contribute to that theory. A finite record reveals behavior within a visited range, while the martingale classification can depend on an unvisited tail. Stochastic volatility adds another connection and another distinction. In Sin's model, positive alignment of price and volatility shocks is central to strictness, but different pricing measures can give the same observed law different valuation implications. These results motivate measuring price–variance association while keeping its statistical and valuation interpretations explicit.

The methodological contribution is a pair of daily and intraday signals built around that association. Both compare positive and nonpositive level dependence using the same background-variance specification. The daily method adjusts for recent market volatility, leaving the stock's own level available to explain additional variation. The intraday method anchors prices within each session and updates its background state between sessions. Their construction addresses a recurring measurement problem: a sufficiently flexible volatility adjustment can absorb the pattern we want to detect.

The empirical contribution is an evaluation of those choices, including simulation sensitivity, calibration, historical alarms, and a supplementary forecasting comparison. In the shared-market-scale simulations and a constant-elasticity diffusion family, the daily statistic detects roughly 39–72% of the specified alternatives, against 7–14% for calibrated GSADF. In historical data, its sparse crisis-cohort alarms coexist with a coherent early signal across D.R. Horton, Lennar, Pulte, and Toll Brothers, all around 400 sessions before their 2005 peaks. The intraday score adds modest out-of-sample drawdown information to GSADF. Part II presents these findings alongside the settings where each method is weaker. Calibration is part of the result: the reference distribution determines what counts as unusual and can materially change the resulting alarms.

1.2 How to assess the evidence

The principal distinction is between a pricing defect under Q and a measured pattern under P. Simulation detection rates have known generating models; historical episode and calm-control labels are comparison groups. Reported calibration and forecast results describe the evaluated procedures and populations. Across those comparisons, GSADF remains the stronger general run-up benchmark, the intraday score contributes incremental drawdown information, and the daily statistic's contribution is its sensitivity and selective historical alarm pattern rather than a demonstrated forecasting gain.

The research used retrospective episodes and current-constituent universes, with overlapping windows and repeated method development. Chronological out-of-sample fits therefore test temporal transfer within this research record, rather than an untouched research holdout. These shared qualifications apply throughout the empirical part. More specific limitations are stated where they change the interpretation of a result.

2 Background and relation to existing work

The literature approaches bubbles through several different observable features: prices relative to cash flows, explosive growth, volatility that increases with the price level, and run-ups followed by crashes. Each approach starts from an economic mechanism and then asks which implications can be measured. The present research connects the strict-local-martingale approach to practical monitoring and to traded contracts. Its closest predecessors therefore include both mathematical valuation papers and empirical detection methods.

2.1 From fundamental value to a pricing defect

In the classical present-value formulation, an asset's fundamental value is the discounted value of its future cash flows. A bubble is an additional price component consistent with the pricing equation but not explained by those cash flows. Empirical tests must consequently specify the cash-flow process and the way investors discount it. Gürkaynak's survey explains why this creates an identification problem: evidence attributed to a bubble can also be explained by changes in fundamentals or discounting [10]. A striking price path alone does not resolve that comparison.

The strict-local-martingale literature provides a precise continuous-time valuation framework. Cox and Hobson study prices that are local martingales under a pricing measure but lose expectation, including local-volatility and feedback models [5]. Jarrow, Protter, and Shimbo place such behavior in a theory of asset-price bubbles and market completeness [14]. In the finite-horizon, discounted setting used here, the object is the defect in (1). This makes the mathematical question explicit: under which models and trading rules does current value exceed expected terminal value under the specified pricing law?

The specification of that law matters in an incomplete market. Herdegen and Schweizer define fundamental value through superreplication and develop a symmetric treatment of assets and numeraires. Their strong-bubble concept differs from a bubble defined under one chosen equivalent local martingale measure; their Example 5.5 explicitly exhibits different martingale behavior under two such measures [12]. Our results concern the fixed-measure defect. The contribution of the identification analysis is to make the implications for observation precise: tail completions and changes of pricing measure can preserve the available price information while changing that defect.

2.2 What volatility can reveal about a bubble

Jarrow, Kchia, and Protter offer the closest methodological starting point for the signals in this manuscript [13]. For a suitable driftless diffusion $\mathrm{d}S_t = \sigma(S_t)\,\mathrm{d}W_t$, strictness is governed by the upper-tail integral

$$\int_c^\infty \frac{x}{\sigma^2(x)}\,\mathrm{d}x.$$

The economic idea is that sufficiently rapid growth of volatility at high prices can generate a pricing defect. Their detection procedure estimates the diffusion coefficient from observed prices and then assesses its tail behavior. They explicitly identify the extrapolation problem and propose parametric and reproducing-kernel methods for extending the estimate beyond the observed range.

Nonparametric diffusion estimation supplies the local measurement tools. Florens-Zmirou studies estimation from discretely observed paths; Bandi and Phillips develop nonparametric methods using occupation-time information [2, 7]. Repeated visits to a price region provide information about volatility there. More frequent observations improve that local information, but need not reveal how the coefficient behaves at prices never reached. Obayashi, Protter, and Yang extend the detection approach toward

dating bubble episodes: they construct daily signals from intraday observations and smooth the noisy classifications with a hidden Markov model [20]. Thus both volatility-based detection and the use of intraday data to produce a daily monitor have clear precedents.

Our theory specifies exactly what remains undetermined when the coefficient is known only on a visited interval, including the lower endpoint that matters for rebalanced wealth. The testing results then turn this observation into a statistical statement: within the stated model classes, a true-martingale completion can approximate the observed law of a strict local martingale arbitrarily well. These are precise formulations of an existing identification concern, with consequences for what assumptions a useful test must impose.

The empirical methods take a more local target. They measure whether *return variance* rises with the price level after accounting for a declared background scale. For intuition, if $\sigma(x) = ax^p$, then return variance is $a^2x^{2(p-1)}$. The positive exponent estimated by a signal is therefore connected to the classical tail mechanism, while remaining an observable feature of the fitted window. Our methodological contribution is the construction of that measurement under changing market volatility, together with its calibration and comparative evidence. It is a different use of the volatility insight from extrapolating a fitted coefficient to infinity.

2.3 Explosive prices, episode dating, and crash risk

Another influential approach asks whether prices exhibit temporarily explosive autoregressive dynamics. In a simple representation, the question is whether the coefficient in $y_t = \rho y_{t-1} + u_t$ exceeds one over some interval. The generalized supremum augmented Dickey–Fuller test scans windows with varying starting and ending dates. This flexibility lets it detect multiple episodes that a test tied to one starting date may miss; the associated backward-window statistics support episode dating [21, 22]. Panel methods can gain information by grouping series with similar explosive behavior [18].

These tests give a natural practitioner benchmark because they formalize the familiar pattern of accelerating prices. Their statistical target is different from the tail condition above: explosive conditional-mean dynamics and price-dependent conditional variance can appear separately. This distinction makes the comparison informative. It lets us ask whether the proposed variance signals detect patterns that an explosive-price statistic tends to miss, and whether those patterns occur in different historical episodes.

Volatility adjustment is already an important part of the explosive-testing literature. Kurozumi, Skrobotov, and Tsarev construct a time transformation to handle nonstationary volatility and recover a pivotal limiting null distribution [17]. Our findings about the calibration of GSADF belong in that context. The empirical comparisons here evaluate the specified GSADF implementation and its calibrated thresholds; they are not comparisons against every volatility-robust extension. The distinctive evidence is the side-by-side behavior of a variance signal and an explosive-price benchmark under common simulation and historical designs.

A further strand studies what happens after observed run-ups. Greenwood, Shleifer, and You find that sharp increases do not generally imply low subsequent average returns, but do imply higher crash probabilities; characteristics such as volatility, turnover, and issuance help distinguish the outcomes [9]. This is the relevant precedent for our supplementary drawdown comparison. The intraday result adds information beyond the GSADF score in the reported forecasting design. Its contribution is that particular incremental comparison, rather than the general proposition that volatility or run-up characteristics can predict crashes.

2.4 Separating price dependence from background volatility

Financial volatility changes over time and responds asymmetrically to returns. Glosten, Jagannathan, and Runkle provide an influential model of that asymmetry [8]. Such dynamics matter here because an observed price–variance association can arise through the joint evolution of prices and a separate volatility state. A useful measurement procedure must specify which variation its background model explains and which variation remains available to the price-level term.

The connection to stochastic-volatility bubble theory is also substantive. In Sin's model, positive alignment of price and volatility shocks can produce strict-local-martingale behavior; the same model also illustrates the role of the pricing measure [24]. Price–volatility association is consequently of interest beyond one-dimensional local volatility. The simulations in this research examine how the signals respond to both structural level dependence and correlated volatility shocks. That response helps explain what the statistic measures across model families.

The daily construction uses a predictable market volatility scale and compares positive and nonpositive price-level dependence with the same nuisance specification. The intraday construction anchors the price within each session and updates the background state between sessions. These choices preserve a distinction between movements inside the measurement window and changes in the background scale. Their value is assessed through simulation, historical alarm patterns, and calibration transfer. This is the central methodological addition to the earlier volatility-estimation approach.

The statistical tools used to assess evidence also have established antecedents. Split-likelihood universal inference supplies a way to construct finite-sample evidence by estimating an alternative on separate data and bounding the null likelihood [27]. The e-BH procedure controls false discovery rate from valid e-values under arbitrary dependence [26]. Rank calibration uses exchangeability to compare a new score with reference scores [25]. We apply these ideas to specified observation models and distinguish their guarantees from the empirical reference calibration used by the practical scores. The findings about how calibration changes sensitivity and historical alarms are part of the contribution, alongside the score designs themselves.

2.5 Investment, rebalancing, and leveraged funds

A bubble in an asset raises a separate question about the wealth generated by trading it. Herdegen and Herrmann connect strict local martingales with a generalized Johansen–Ledoit–Sornette crash model and solve a power-utility investment problem. Under their assumptions, investors with relative risk aversion above one continue to hold the bubble when instantaneous expected returns are positive [11]. This gives a useful investment perspective: the existence of a bubble does not by itself determine an optimal position. Their crash model also connects the strict-local-martingale literature to the separate tradition of modeling log-periodic behavior before crashes.

Our investment question concerns a prescribed rule: keep a constant fraction of wealth in the asset. The tools are established martingale and diffusion criteria. Mijatović and Urusov give the stochastic-exponential criterion we apply, while Karatzas and Ruf analyze the auxiliary drift family that arises in the calculation [15, 19]. The result developed here classifies the exposure rule across both price endpoints. In particular, continuous buying into an asset approaching zero can create strictness in wealth even when the asset is a true martingale. The comparison with finite trading makes the economic importance of that boundary mechanism concrete.

For leveraged exchange-traded funds, Avellaneda and Zhang establish the path dependence of multi-period returns and their relationship to realized variance, financing, and expenses [1]. That work explains

why a fund's cumulative return differs from a fixed multiple of the underlying's cumulative return. Part III addresses a complementary issue: how finite rebalancing, nonnegativity, and liquidation affect the martingale classification. The empirical tracking-cost evidence then places those idealized wealth processes beside observed funds. The established variance-drag explanation and the endpoint classification answer different questions about the same trading rule.

2.6 What options and constituents add

The option-pricing consequences of strict local martingales are already central to Cox and Hobson: familiar put–call parity and European–American relations can change, and admissible trading strategies affect valuation [5]. Carr, Fisher, and Ruf develop a joint-superreplication framework that produces a parity-restoring price [4]; Bayraktar, Kardaras, and Xing study related strict-local-martingale and American-option phenomena [3]. These papers supply the valuation background for Part IV.

Our contribution in that part is to organize the information supplied by actual contract observations. A quote becomes informative about the defect only after its pricing convention and exercise rights have been specified. The resulting inequalities distinguish an expectation-priced European option, a parity-restored value, and an American exercise value. This provides a practical way to read option evidence and to understand why different contract conventions can reveal or absorb the same defect.

Constituent information offers another route beyond an aggregate price. The underlying change-of-measure and explosion ideas come from the strict-local-martingale literature, including Kardaras, Kreher, and Nikeghbali [16]. We use them to express constituent defects through weights stopped at the basket's lifetime and their conditional expectations. The useful addition is an explicit attribution formula and an account of the extra information its recovery requires. Together, the fund, option, and constituent results extend the question from detecting a feature of one price series to understanding the exposures and contracts an investor actually holds.

Part I

Theory and the Limits of Identification

When does a strategy inherit a bubble, and what can observed prices determine?

3 Constant exposure and the two endpoints

Consider a discounted asset satisfying

$$\mathrm{d}S_t = \sigma(S_t)\,\mathrm{d}W_t, \qquad S_0 = s_0 > 0, \tag{2}$$

under Q. The coefficient is Borel and nowhere zero on $(0,\infty)$, with $1/\sigma^2$ integrable on every compact subinterval. The nonnegative solution is absorbed at zero. These assumptions allow weak diffusion laws; differentiability of the coefficient is unnecessary for the criterion below.

An investor maintains exposure $\pi > 0$ to the asset and the rest of wealth in cash. Before the asset reaches zero, self-financing gives $\mathrm{d}V_t^\pi = \pi V_t^\pi \mathrm{d}S_t/S_t$. The wealth under study is defined at all times by

$$V_t^\pi = v_0 \mathscr{E}\left(\pi \int_0^\cdot \frac{\mathrm{d}S_u}{S_u}\right)_t. \tag{3}$$

The exponential is continued at exit according to the convention in the martingale criterion of Mijatović and Urusov [19]: it is set to zero when its quadratic-variation integral diverges at exit. We call this the *wealth representation*. It is the specified model of wealth, not a proof that every implementable strategy through default has this value. In this driftless diffusion the integral diverges when S reaches zero, so this wealth is then zero.

For any fixed $c > 0$, define

$$I_0 = \int_0^c \frac{x}{\sigma^2(x)}\,\mathrm{d}x, \qquad I_\infty = \int_c^\infty \frac{x}{\sigma^2(x)}\,\mathrm{d}x. \tag{4}$$

Local integrability makes their finiteness independent of c. The established driftless criterion says that S is strict on every positive finite horizon exactly when $I_\infty < \infty$; the corresponding lower-endpoint test says that zero is accessible exactly when $I_0 < \infty$ [15, 19].

Theorem 1 (Constant-proportion classification)**.** *Under* (2)*, the stated coefficient assumptions, and the wealth representation* (3)*, for every finite $T > 0$,*

$$V^\pi \text{ is strict on } [0,T] \quad \Longleftrightarrow \quad \left(0 < \pi < \tfrac{1}{2} \text{ and } I_0 < \infty\right) \text{ or } \left(\pi > \tfrac{1}{2} \text{ and } I_\infty < \infty\right). \tag{5}$$

In particular, $\pi = 1/2$ always gives a martingale on each finite horizon.

The result is conditional on the cited martingale criterion and the wealth representation. Its deterministic endpoint calculation is the project's proof. The auxiliary-diffusion family already appears in Karatzas and Ruf, equation (7.4) [15]; the contribution here is the classification across exposure and its implications for rebalancing and default.

Exposure	True martingale on every finite horizon iff	Endpoint that matters
$0 < \pi < 1/2$	$I_0 = \infty$	Zero
$\pi = 1/2$	Always	Neither permits auxiliary exit
$\pi > 1/2$	$I_\infty = \infty$	Infinity

Table 2: The exposure classification under the stated wealth convention.

3.1 Why one-half appears

The criterion examines an auxiliary diffusion with drift $\pi\sigma^2(x)/x$ and the original diffusion coefficient. Its scale density is $(x/c)^{-2\pi}$. Above one-half, the upper endpoint of its scale is finite; below one-half, the lower endpoint is finite. The corresponding Feller integrands reduce to

$$\frac{1}{2\pi - 1}\frac{x}{\sigma^2(x)} \quad (\pi > 1/2), \qquad \frac{1}{1 - 2\pi}\frac{x}{\sigma^2(x)} \quad (0 < \pi < 1/2). \tag{6}$$

At one-half the scale is logarithmic and neither endpoint allows this exit. Neither endpoint is "good" in the terminology of the cited criterion, so auxiliary exit decides strictness. Appendix B gives the calculation in detail.

The lower endpoint changes the usual inheritance intuition. Below one-half, continuous rebalancing into an asset that can reach zero creates strict wealth even when the asset itself is a martingale. The investor buys more units as the price declines, continually replenishing the risky fraction from cash. The model rides this process all the way to zero wealth. That pathwise description is one part of the exposure classification, which also accounts for gains on surviving paths through their expectation under Q.

Example 2 (Two reversals at a quarter allocation)**.** For absorbed Brownian motion, $\sigma(x) = 1$, $I_0 < \infty$ and $I_\infty = \infty$. The asset is a martingale but $V^{1/4}$ is strict. For $\sigma(x) = x^2$, $I_0 = \infty$ and $I_\infty < \infty$. The asset is strict but $V^{1/4}$ is a martingale. Part III shows that every finite rebalancing grid reverses the continuous-wealth classification in both examples.

At $\sigma(x) = 1 + x^2$, both integrals are finite. Every positive exposure except one-half then produces strict wealth. The boundary behavior, as well as the exposure, determines the outcome.

Research record: claim 001; experiments 001 and 006.

3.2 Long horizons, changing exposure, and independent assets

Three extensions help delimit the classification. Applying the infinite-horizon criterion of Mijatović and Urusov, constant nonnegative exposure gives a *uniformly integrable martingale over all time* only at $\pi = 0$ [19]. Here uniform integrability is required together with the martingale property. For example, geometric Brownian wealth at positive constant exposure is a martingale on each finite horizon but fails uniform integrability over all time.

Changing exposure can restore uniform integrability. For geometric Brownian motion $\sigma(x) = \kappa x$, with $\kappa > 0$, a bounded deterministic schedule $\pi(t)$ gives a uniformly integrable martingale exactly when $\int_0^\infty \pi(t)^2 \, \mathrm{d}t < \infty$. For example, $\pi(t) = (1 + t)^{-\gamma}$ qualifies exactly when $\gamma > 1/2$. For an adaptive schedule, the relevant almost-sure energy condition is under the auxiliary measure in Ruf's criterion, not simply under Q [23]. Bounded exposure alone is insufficient for a general local-volatility asset.

If two full asset–wealth systems are independent under Q and their product is a continuous local martingale in the common filtration, the product wealth has multiplying mean factors. Writing their normalized defects as δ_1 and δ_2, the product defect is

$$\delta_{12} = \delta_1 + \delta_2 - \delta_1\delta_2. \tag{7}$$

It vanishes exactly when both defects vanish. An independent true-martingale factor does not remove the strictness of another factor. Independence of the full systems is the key assumption behind this factorization.

Research record: claims 074–076; numerical checks summarized in the original paper's Section 3.1.

4 What prices identify

The classification depends on what happens near zero and infinity. A finite record occupies a finite price range. Better measurement inside that range does not by itself supply the missing behavior outside it. This is the extrapolation problem already recognized by Jarrow, Kchia, and Protter [13].

Proposition 3 (Compatible tails)**.** *Suppose σ is known exactly on $(0, R]$, with $c < R$. Set $J = \int_c^R x/\sigma^2(x)\,\mathrm{d}x$. Over all admissible coefficients agreeing with σ on that interval, the upper-endpoint integral ranges over $(J, \infty]$. Its finiteness can take either value. If σ is positive and continuous, the compatible coefficients can be chosen positive and continuous. If only $[r, R]$, $r > 0$, is known, the lower and upper unknown tails can be varied independently.*

Proof. Keep σ unchanged below R and set $g_p(x) = \sigma(R)(x/R)^p$ above it. For $p = 1$ the tail integral diverges; for $p > 1$ it equals

$$\int_R^\infty \frac{x}{g_p(x)^2}\,\mathrm{d}x = \frac{R^2}{2(p-1)\sigma(R)^2}.$$

Varying p yields every strictly positive finite tail contribution, together with infinity. A similar completion below r varies the lower tail. The known middle contribution is unaffected. □

This proposition grants much more than actual observations: exact knowledge of a coefficient on an interval. It is an identified-set statement, not a confidence interval or a finite-sample testing theorem. The latter requires comparing observation laws.

Theorem 4 (Finite-horizon testing obstruction)**.** *Fix a positive finite horizon and finitely many observed price paths. In the class of driftless local-volatility diffusions described above, including $P = Q$, every bounded randomized test has power at each strict alternative no larger than its rejection probability maximized over all true-martingale nulls. The conclusion persists when coefficients are required to be positive and continuous. It also has a version allowing equivalent physical measures.*

For $P = Q$, the argument is particularly transparent. Replace the coefficient above a large level n by a linear tail. The completed price is a martingale and agrees in stopped law with the alternative until the price reaches n. The probability of reaching that level before the horizon tends to zero, so the observed path laws become arbitrarily close. A finite number of paths does not change the argument. Weak existence and stopped uniqueness in law are cited diffusion inputs; their full reconstruction is not part of the project. Appendix B records the probability comparison.

The theorem concerns *uniform* size over a broad null class. Restricted models can support informative tests by excluding the compatible completions, for example through a specified tail law. Under $P = Q$, a

known full law identifies $\mathbb{E}_Q[S_T]$; the obstruction concerns inference from finite records uniformly across the admitted laws.

Research record: claims 002, 003, 006, and 013.

4.1 Sin's model: feedback and the pricing measure

Sin's stochastic-volatility model provides a second theoretical reason to study positive price–volatility association [24]. Under a specified pricing probability Q^s, write its dynamics as

$$\mathrm{d}S_t = S_t v_t^{\alpha} \gamma \cdot \mathrm{d}W_t, \qquad \mathrm{d}v_t = v_t a \cdot \mathrm{d}W_t + \kappa(L - v_t)\mathrm{d}t, \tag{8}$$

where W is two-dimensional Brownian motion, $\alpha, \kappa, L > 0$, a, γ are nonzero constant vectors, $S_0 > 0$, and $v_0 = 1$. The price and volatility factor remain positive. Sin's Theorem 3.2 says that S is a martingale on all finite horizons exactly when $a \cdot \gamma \leq 0$. Positive alignment therefore produces strictness on some finite horizon.

The dot product is also the sign of the instantaneous association between price and volatility shocks. When it is positive, upward price shocks arrive with upward volatility shocks. Under the auxiliary change of measure used to examine the price's expectation, that alignment adds an upward term of order $v^{1+\alpha}$ to the volatility drift. The resulting explosion mechanism creates the expectation shortfall. This is a stochastic-volatility route to a bubble, alongside the local-volatility route in which the diffusion coefficient grows rapidly with the price.

That connection changes how a signal should be interpreted. A positive fitted level exponent may summarize explicit local-level dependence or correlated volatility shocks. Those are different explanations of the coefficient, but both warrant attention when studying bubble-related price–variance behavior. The simulation response to correlation in Part II is informative about that broader sensitivity. The model must still be specified to turn the association into a strictness criterion: Sin's multiplicative volatility dynamics differ from the square-root variance dynamics used in those simulations.

Sin's model also separates this mechanism from identification of a pricing measure. With nonparallel vectors, $|a \cdot \gamma| < |a||\gamma|$, his Theorem 3.6 supplies an equivalent measure under which the same price is a martingale on any specified finite horizon. Taking $P = Q^s$ and choosing a horizon with strictness gives one physical law compatible with both positive and zero pricing defects. The change of measure alters the volatility drift while preserving the instantaneous covariance structure. Thus covariance information helps characterize the observed feedback, while a valuation conclusion additionally requires a pricing law. The two roles of Sin's model—a positive feedback mechanism and pricing-measure ambiguity—belong together.

Research record: claim 003, part 3; Sin's Theorems 3.2 and 3.6, as recorded in AX-05.

4.2 What is positively identified inside the range

There is a useful positive result. For a continuously observed path of the diffusion, let L_T^x be semimartingale local time, normalized by the occupation identity $\int_0^T f(S_t)\sigma^2(S_t)\,\mathrm{d}t = \int f(x) L_T^x\,\mathrm{d}x$. With zero contribution when the denominator is zero, the visited-range identity is

$$\int_0^T \frac{S_t}{L_T^{S_t}} \mathbf{1}_{\{S_t>0, L_T^{S_t}>0\}}\,\mathrm{d}t = \int_{(\min S, \max S)\cap(0,\infty)} \frac{x}{\sigma^2(x)}\,\mathrm{d}x. \tag{9}$$

It is an application of the occupation formula and positivity of local time on the visited interior. The identity is pathwise and transfers under equivalent physical measures. It identifies the visited portion, not

its unobserved completion. Estimating it from sampled prices introduces another problem, especially near levels with little occupation.

Under a declared common power tail, pooling names expands the union of visited price levels. Claim 015 gives a limiting detection probability at least as large as that for any one name, without assuming independence. This is a coverage gain; no estimation-rate improvement is asserted. Separate reviewer simulations of the later pooled likelihood constructions in claims 062–063 found no improvement over per-name testing.

Research record: claims 008, 012, 013, and 015; experiment 013 for sampled visited-integral estimation; reviewer simulations in claims 062–063 for pooling.

4.3 Why sampling frequency is not a universal cure

In a separable model for log price X, suppose its instantaneous variance is $v_t = \exp(a(t) + \beta X_t)$ and a is deterministic, continuous, and of finite variation. Continuous observation yields

$$\beta = \frac{[\log v, X]_T}{[X]_T}, \qquad v_t = \frac{\mathrm{d}[X]_t}{\mathrm{d}t}. \tag{10}$$

Finite variation contributes no quadratic covariation, so this follows by polarization. Stochastic volatility with nonzero quadratic variation does not satisfy that premise. Nor does (10) provide a noise-robust estimator of the derivative of quadratic variation from daily prices.

Under $P = Q$ and the cited exact price laws, on every fixed finite grid the normalized laws of $\mathrm{d}S = \kappa S\,\mathrm{d}W$ and $\mathrm{d}S = \kappa S^2\,\mathrm{d}W$, started at one, approach one another in total variation as $\kappa \downarrow 0$. The first is a martingale and the second strict, with a fixed separation in elasticity. Thus no uniform finite-grid power or required-frequency guarantee follows from elasticity separation and a time-variation budget alone. A lower bound on the relevant variance scale is missing. At a given scale, richer intraday observations can still provide the substantial gains documented in the next part.

Research record: claims 071–073.

Part II

Price-Level Volatility Signals

Proposed daily and intraday methods, their calibration, and what they do on real prices.

5 The signal and the background variance

The empirical target is a positive dependence of return variance on the stock's price level after accounting for background volatility. The motivation comes from a constant-elasticity diffusion $\mathrm{d}S = \kappa S^p \mathrm{d}W$. Its instantaneous log-return variance is $\kappa^2 S^{2(p-1)}$. We therefore use the convention

$$\beta = 2(p-1) \tag{11}$$

throughout this manuscript. A positive β is the direction associated with strictness in that global, time-homogeneous diffusion. On observed records it supplies a local statistical signature whose strength can be estimated and compared across stocks and periods.

Background volatility is the main difficulty. A stock can rise during a market-wide volatility expansion without its variance depending structurally on its own price. Conversely, a state fitted flexibly to that stock's own recent returns may absorb a genuine level effect. Our proposed methods make different choices about which variation belongs to the background and which remains available to identify the level signature.

5.1 One conditional-return model on both sides

Let $R_t = \log(S_t/S_{t-1})$ and let x_{t-1} be log price relative to a declared anchor. Write

$$R_t = \mu + \sqrt{v_t(\beta)}\, Z_t, \qquad v_t(\beta) = A_t \exp(\beta x_{t-1}), \tag{12}$$

where Z_t has a Student-t law scaled to variance one and degrees of freedom $\nu > 2$. The same location, background-scale rule, and innovation family enter both sides. The contrast is between $\beta > 0$ and $\beta \le 0$; including negative values allows variance to increase as the price falls.

This conditional-return model is fitted under the observation law. Stochastic-volatility processes provide separate stress cases for its interpretation and calibration.

Let $\ell_n(\beta)$ be the accumulated log density over a scored record, with nuisance quantities fixed or updated by their predeclared rule. A batch contrast is

$$A_n^{\mathrm{batch}} = \sup_{\beta \ge 0} \ell_n(\beta) - \sup_{\beta \le 0} \ell_n(\beta). \tag{13}$$

The implementation maps it to $\frac{2}{\pi} \arctan \max(0, A_n^{\mathrm{batch}})$. This bounded score has many ties at zero, denoting no positive evidence under the fitted contrast.

For monitoring, an equal-weight mixture over $\mathscr{B} = \{1/8, 1/4, 1/2, 1, 2, 4, 8, 16\}$ supplies the numerator:

$$L_n = \log\left(\frac{1}{8}\sum_{b\in\mathscr{B}} e^{\ell_n(b)}\right) - \sup_{\beta\le 0}\ell_n(\beta). \tag{14}$$

The mixture's predictive weights evolve using previous returns. The null fits one common exponent to the whole prefix; it is not a product of separately maximized one-step likelihoods. Reference records are scored over their full scheduled horizons, and the reference distribution of the path maximum sets the alarm cutoff.

Both profiles are numerical searches. Failures and unresolved profiles have explicit statuses, are retained in denominators, and do not create positive evidence. The real-data implementation is an empirically calibrated statistic, not a certified sequential e-value. Appendix A gives the density, parameter domains, numerical conventions, and their relationship to the separate exact-model guarantees.

5.2 Daily market-scale construction

For daily observations, anchor $x_{t-1} = \log(S_{t-1}/S_0)$ at the close immediately preceding a launch. Use the last 20 completed SPY returns to form

$$M_t = \frac{1}{20}\sum_{j=1}^{20}(R^{\mathrm{SPY}}_{t-j})^2, \qquad m_t = M_t/M_{\mathrm{ref}}, \qquad A_t = w m_t^g. \tag{15}$$

Here M_{ref} is the geometric mean of the scale over the prior 252-session training period, frozen at launch. The background market scale advances using completed market sessions. There is no stock-specific moving variance recursion inside the scored window.

Fit (μ, w, g, ν) under $\beta = 0$ on the training returns, then freeze them and score the following 252 returns. The batch statistic and the daily monitor have separate calibrations. The stock's training data estimate the constant nuisance parameters; the moving background state comes from SPY. A stock in SPY contributes to its own proxy, with that dependence most direct when scoring SPY itself.

This construction asks whether positive own-level dependence remains after one predictable market-scale adjustment. Independent stock-specific volatility remains a source of residual variation, examined explicitly in the simulations.

Research record: experiments 055–056 and the D37 statistic design.

5.3 Intraday session-anchor construction

The intraday method uses regular-session five-minute closes. At each monthly launch it estimates a diurnal clock from the preceding 252 sessions and fits $(\mu, h_{\mathrm{init}}, \nu)$ at $\beta = 0$. Let c_j denote the clock for slot j, normalized to mean one. Within session d, use

$$v_{d,j}(\beta) = h_d c_j \exp\left(\beta \log\frac{S_{d,j-1}}{s_d}\right), \tag{16}$$

where s_d is the session's first bar close. The state h_d remains fixed inside the session. After a completed session it updates from the mean clock-adjusted squared residual, with a fixed 20-session half-life. The state rule and fitted nuisance parameters are common to every exponent.

The level anchor resets each session while the likelihood evidence accumulates across 21 sessions. A full launch has 1,617 adjacent five-minute returns. Overnight returns are excluded. Missing required

observations end the supported prefix; early closes follow the declared schedule. This method preserves within-session dependence that a variance state updating at every bar could absorb. It can still absorb slower effects between sessions, and the state can be misspecified during abrupt changes.

The original intraday implementation stores the exponent of standard deviation, half the variance exponent used here. Its support $\{1/16, \dots, 8\}$ is therefore the same as $\mathscr{B}$ in (14). This notation change does not alter the algorithm.

Research record: experiment 054; experiments 052–053 for its diagnostic motivation.

5.4 Why the earlier likelihood score was replaced

An earlier split-likelihood procedure selected an exact diffusion forecast from a training catalogue and divided its likelihood by a drift- and scale-profiled Gaussian null. It had an honest in-model guarantee, but its real-data silence was not a clean demonstration of a fundamental power barrier. In 2,287 of 3,000 broad-universe diagnostic windows, the forecast was itself a null member, so the ratio could not exceed one. A training-selected scale from a three-point grid also competed against a scale fitted on the evaluation record. Scale misfit dominated the negative score in the cases examined.

The diagnosis also found weak average daily separation even when fitting on the evaluation record. A free-exponent likelihood contrast exceeded log 20 on 12.1% of consensus positives and 11.7% of calm controls. Monthly realized-variance regressions controlling for market variance gave pooled level elasticities of +0.05 (SE 0.04) on run-ups and +0.02 (SE 0.08) on consensus positives, compared with −0.27 on unselected windows and +0.59 on calm controls. The construction problem and the weak average signature were both part of the diagnosis.

The subsequent designs therefore changed how they isolate the pattern. A shared Student-t model repaired the likelihood comparison; the daily market scale and intraday session anchor then preserved variation that an own-stock volatility state could absorb. The resulting signals have measurable simulation sensitivity and distinctive historical behavior, reported below.

Research record: experiments 049–051 for the diagnosis and initial repair; 054–056 for the later constructions.

6 Calibration is part of the method

A score ranks records according to a particular contrast. A threshold turns that ranking into an operational decision. Two procedures can produce similar rankings and very different alarm frequencies because their reference distributions differ. Conversely, forcing both to flag 5% of one reference sample does not establish that they make the same errors on the next population.

6.1 The strict rank rule and its premise

For m reference scores $Z_1, \dots, Z_m$ and target score Z_*, let

$$k = \lceil (m+1)(1-\alpha) \rceil, \qquad q = Z_{(k)}, \qquad \text{alarm if } Z_* > q, \tag{17}$$

with $q = +\infty$ when $k > m$. Under exchangeability of the selected references and target, this rule has marginal rejection probability at most α. Strict exceedance preserves conservatism in the presence of ties. For $m = 499$ and $\alpha = 0.05$, the cutoff is the 475th ordered reference score. An additional zero floor can only reduce rejection.

This is a standard rank-calibration argument, applied to the entire score construction [25]. For the monitor, the score used in calibration is a complete record's maximum, so the controlled event concerns at

least one crossing during that declared horizon. It does not concern indefinite monitoring, every monthly relaunch, or all stocks jointly.

The difficult premise is the score law *after selection.* Choosing references solely from the past prevents a form of look-ahead but does not make calm references exchangeable with all future targets. Fitting nuisance parameters, excluding missing records, choosing a market proxy, and changing window lengths are all part of that law. Claim 045 supplies conditional extensions with explicit distributional-error budgets, but those budgets have not been estimated for the market replays here. Past-only selection is insufficient (claim 045), as is stationarity alone (the separate counterexample in the refuted-research record).

Research record: claim 045; `claims/refuted/d34-stationarity-calibration.txt`.

6.2 A measured calibration effect on GSADF

The lab first calibrated GSADF using 15 daily calm windows (103 name-records) and 14 hourly calm windows (98 name-records). Each statistic was converted to a simulated-random-walk p-value at the record's length. Empirical thresholds were chosen from the calm records. Each calm window was then evaluated against a threshold fitted on the other windows; positive episode records used the threshold from all calm records.

Records	Conventional	Held-out calm	95% interval	Episodes
Daily, full record	21.4	6.8	1.9–12.5	63.6
Daily, to peak	19.4	6.8	1.0–14.3	59.4
Hourly	12.2	7.1	2.0–13.3	76.2

Table 3: GSADF flag rates (%) at a 5% reference target, before and after empirical calibration. "Conventional" is the calm-record rate under the simulated-null cutoff. "Episodes" uses the empirical cutoff. Intervals resample whole calm windows; they are not independent-trial intervals. Source: experiment 040.

Calibration substantially reduces calm-record flags while retaining many episode flags: the daily full-record calm rate falls from 21.4% to 6.8%, with 63.6% of episode records still flagged. The historical experiment used interpolated quantiles and a tie-inclusive p-value rule, not precisely (17); later methods use the conservative rank rule. The daily to-peak comparison uses an outcome-selected endpoint; the full-record comparison avoids that particular selection.

Applying the daily cutoff unchanged to the broad universe reveals another dimension. At 13-, 24-, and 72-month lengths, conventional GSADF flag rates of 15.3%, 19.0%, and 33.2% become 8.2%, 10.9%, and 23.0%. In 2009–19, the corresponding calibrated rates are 6.4%, 7.4%, and 14.8%. A threshold fitted to shorter calm records does not transfer evenly to six-year windows, even after converting the statistic to a random-walk p-value. These are rates on unlabelled records, not estimates of size.

Research record: experiments 039, 040, and 045; 416,380 overlapping name-windows.

6.3 Exact-model guarantees and empirical calibration

The research contains exact-model e-values as well as the empirical scores. For a normalized forecast density q and a null likelihood envelope $\sup_{\theta\in\Theta_0} p_\theta$, the split-likelihood ratio is an e-value when the true null density is included and the forecast is properly conditioned on training information. This is an application of universal inference [27]. e-BH can then control false discoveries across names under arbitrary dependence [26]. Appendix B gives the argument.

The distinction is especially consequential for monitoring. Predictable forecasts with a coherent whole-record null and certified likelihood bounds support the lab's anytime-valid construction. The real-data Student-t monitors instead freeze estimated nuisance parameters and use floating-point profiles; the stated exact null premise and numerical certification are not available. Their evidence is calibrated empirically as a statistic.

Known approaches to explosive-root testing already address changing volatility, including time transformation [17]. The lab's empirical calibration of GSADF is one benchmark implementation. The comparisons below are against that implementation, rather than the full family of volatility-robust tests.

7 Simulation: recovery, power, and confounding

The daily statistic has substantial sensitivity to positive level dependence in its intended family. In the shared-market-scale and constant-volatility CEV alternatives, its batch detection rate is 38.6–72.2%, against 7.4–13.8% for calibrated GSADF. These comparisons use matching null calibrations and the same paths. They measure sensitivity to the specified level alternatives; the methods' native null hypotheses differ.

7.1 The daily market-scale method in its intended family

The daily validation used 20 market warm-up sessions, 252 training sessions, and 252 scored sessions. Each null cell had 499 calibration and 500 evaluation paths; each alternative had 500 evaluation paths. Methods used the same generated observations. The fitted procedure was compared with known nuisance parameters where defined, with its market exponent fixed to zero, and with calibrated GSADF. Alternative cutoffs came from the matching zero-level-effect family.

Generating model	Batch	Monitor	No-market batch	GSADF
Market scale, $\nu = 6, \beta = 1$	38.6	30.6	18.6	7.4
Market scale, $\nu = 6, \beta = 2$	66.0	60.2	40.8	11.2
Market scale, $\nu = 10, \beta = 1$	38.6	29.4	14.2	8.4
Market scale, $\nu = 10, \beta = 2$	71.0	58.2	32.0	13.8
Persistent effect, $\beta = 1$	36.8	31.2	22.0	10.6
Constant-volatility CEV, $p = 2$	72.2	60.8	73.8	9.6
Market-driven SV, $p = 2, \rho = 0$	23.2	23.0	11.6	10.8
Own-stock SV, $p = 2, \rho = 0$	10.4	10.0	10.6	10.4

Table 4: Selected simulation flag rates (%) under matched calibration. SV denotes stochastic volatility. The shared-market rows include both innovation choices, $\nu = 6$ and 10; the persistent row has the effect in training and scoring. GSADF is a different-target comparator. Source: experiment 055.

The daily statistic recovers a known exponent in its own family. All estimates in the six main recovery cells are finite; median relative errors are at most 6%. Eleven of twelve fitted/oracle recovery checks pass. The failure is the fitted persistent-$\beta = 1$ cell: 79.2% of estimates are positive against a declared 80% target, while the known-nuisance version passes. Absorption of a pre-existing level effect by the $\beta = 0$ prior fit is a plausible explanation for that shortfall.

Across 22 null cells, matched rates average 4.85% for the fitted batch statistic and 4.61% for the monitor. Four exploratory per-cell checks fail across the fitted, oracle, and no-market variants, and an extreme-

negative- exponent stress cell is unresolved because of numerical failures. These rates describe the realized calibration cutoffs.

The power gains are substantive in the intended family. Using the market scale instead of setting $g = 0$ improves sensitivity in market-driven cells. It adds little when the stock's volatility evolves independently of that market. At $p = 2$ with own-stock stochastic volatility, monitor sensitivity is only about 10–20% across the correlations tested. An external scale helps when it tracks the nuisance variation actually present.

7.2 What a positive fitted exponent can mean

In the own-stock stochastic-volatility simulations, a structural $\beta = 0$ with positive price–volatility-driver correlation produces a median fitted exponent of +2.54; negative correlation produces −2.58. There is no true level term in either case. The statistic thus responds to correlated stochastic volatility as well as to explicit local-level dependence. This broader response is relevant to the positive-feedback mechanism in Sin's model (Section 4.1). It also means that the fitted exponent measures an association shared by several mechanisms. The square-root variance process used here is different from Sin's multiplicative factor; its correlation response is evidence about the statistic, not a direct application of Sin's strictness theorem.

The intraday estimator validation reaches a similar conclusion. With a known level effect and independent volatility shocks, the within-day estimator recovers its sign and approximate size. With structural $\beta = 0$ and driver correlation −0.7, its median is about −4.1. Estimators using the stock's own smooth state attenuate a simulated positive effect more than the within-day estimator. Together these experiments explain what the session anchor preserves: a within-day price–variance association that faster own-stock state updates can attenuate.

Research record: experiments 053 and 055.

7.3 Calibration transport can dominate the result

The same daily statistic behaves very differently when calibrated on a different variance law. Figure 1 applies the cutoff from the baseline shared-market, $\beta = 0$ family to other cells with no structural level term. With own-stock stochastic volatility and positive correlation, the monitor flags about 70%, despite being near 5% when calibrated on the matching family. Regime-switching controls similarly produce large increases in the full validation.

This result explains why calibration is part of the method. Matched calibration can absorb the distribution of a nuisance pattern, but the price of doing so may be weaker separation from the alternatives. The relevant empirical question is how closely the reference library resembles the target population in the features that drive the score.

The diffusion simulations were also compared at 10 and 50 substeps per day using coupled increments. Fitted coefficients changed little; a later check found four monitor alarm changes among 2,100 paired paths at fixed coarse cutoffs, indicating relatively stable decisions at those resolutions.

7.4 What the earlier exact-model experiments establish

Exact constant-elasticity likelihoods offer a complementary benchmark. In a two-exponent, four-scale catalogue, simulated null rejection was at most 0.1% at a 5% target. Against $p = 2$, daily power rose from roughly 24–46% over one year to 96–98% over five years; five-minute sampling gave nearly complete detection in the tested settings. This establishes that level dependence can be statistically visible under strong model restrictions.

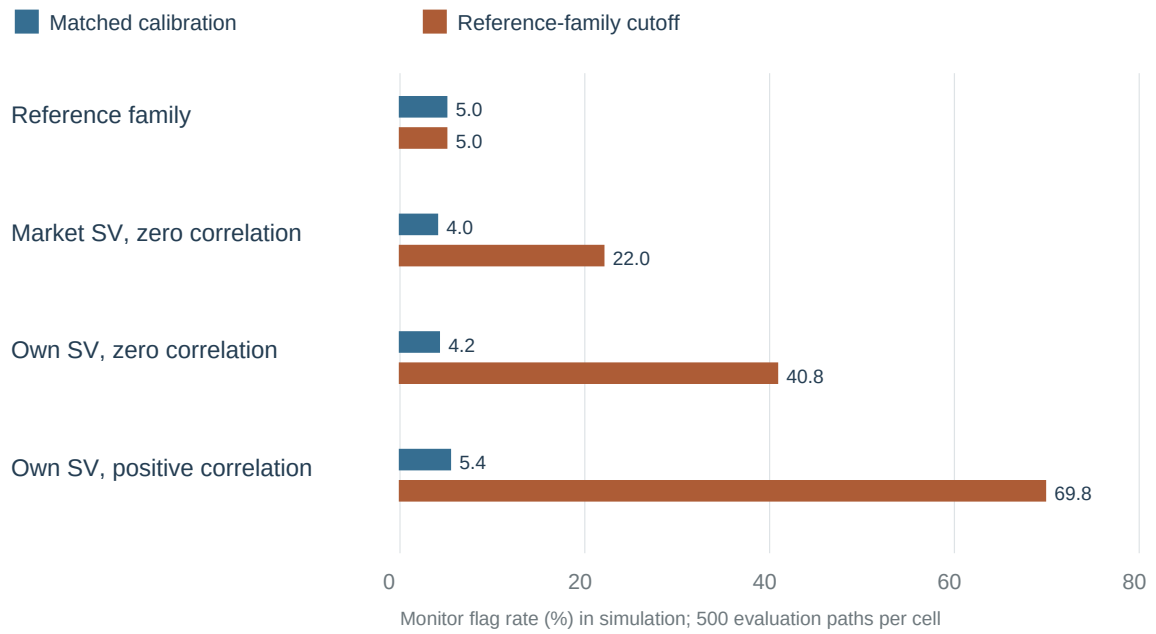


Figure 1: The same daily monitor under matched and transported calibration. "No level term" is the structural simulation label; it does not claim that all these models have the same pricing-martingale classification. Source: experiment 055.

Misspecification changes the picture. A geometric Brownian scale between catalogue points was flagged 24% of the time in a five-minute experiment. Allowing a continuous Gaussian scale fixed that particular problem, but repeated variance regimes broke the expanded procedure on daily panels. Permitting switch dates restored validity within the switch model while reducing power: in one five-year $p = 1.5$ comparison, 44.7% became 17.3% with a single admitted midpoint and 2.1% when the middle 80% of dates were admitted. The experiment quantifies the sensitivity lost as that particular comparison model becomes more flexible.

At high frequency, a declared intraday clock can preserve substantial in-model power. Rounding and observation noise require their own likelihood treatment. The lab's exact hidden-price filter avoids a worst-case penalty at every rounded observation; certified calculations were within roughly one log-likelihood unit of the ideal on simulated \$5 geometric-Brownian null records of 30 and 125 days at cent ticks. This is a result for that observation model, separate from the empirical Student-t replays.

Research record: experiments 019, 022, 025–028; claims 033–038, 055, and 059. See the research map for the distinction between experiments and formal results.

8 Daily observations in the market

The daily evidence has three roles: comparing historical episodes with calm records; observing alarm behavior in a broader population; and checking whether different constructions continue to differ after calibration. Forecasting later outcomes is an additional exercise, considered after the alarm results.

8.1 Data and comparison design

The early historical collection includes the dot-com period, financial and housing-related names around 2007, and later run-ups. Separate records add collapsed names where free data could be recovered. The

broader study uses the pinned current S&P 500 constituents, with 416,380 overlapping windows of 13, 24, or 72 months in the earlier analysis. It is unselected with respect to the score, but not a historical point-in-time investment universe.

Missing delisted names can change results in either direction. Failed firms remove collapses; acquired firms can remove run-ups ending in takeover premiums. The lab recovered only part of a union of public deletion lists, with especially poor historical coverage before 2008. The direction of survivorship bias is therefore ambiguous. Adjusted-close replays also use a retrospective corporate-action record.

The daily market-scale replay uses monthly launches, 252 prior returns, and a 252-session scored suffix. It starts in 1994 because SPY history and the initial scale/training requirements constrain earlier dates. At each launch, references are earlier completed universe records selected by a calm-start predicate on training volatility and drawdown. Up to 499 are taken in a fixed score-blind order, at most one per name per quarter. The batch statistic, monitor maximum, and GSADF maximum receive separate rank cutoffs from those same references.

8.2 The daily market-scale signal changes the alarm pattern

Among 150,519 universe launches with a finite calibration cutoff, the batch statistic flags 2.91%, the monitor 3.05%, and GSADF 6.86%. Including 3,197 early launches without a usable reference gives 153,716 launches and the slightly lower rates used in Table 5. Keeping these denominators separate prevents the absence of a cutoff from appearing to be successful non-rejection.

Launch period	Launches	Batch	Monitor	GSADF	Calm batch
1994-99	20,760	6.3	6.1	8.2	8.9
2000-02	12,516	1.8	1.8	7.1	0.0
2003-07	22,737	4.3	4.1	4.5	4.9
2008-09	9,754	0.9	1.1	5.9	0.2
2010-19	53,715	2.0	2.3	6.2	2.8
2020-21	11,560	1.4	1.9	14.7	1.5
2022-26	22,674	2.4	2.7	4.9	2.3

Table 5: Daily real-data flag rates (%). “Calm batch” restricts to calm-start records. Periods classify the launch, with the following year scored. Early no-reference launches remain in this table. Source: experiment 056.

The daily signal remains sparse in the three turbulent launch cohorts. Batch rates are 1.8% in 2000–02, 0.9% in 2008–09, and 1.4% in 2020–21; monitor rates are 1.8%, 1.1%, and 1.9%. GSADF rates are 7.1%, 5.9%, and 14.7%. The daily procedure’s response therefore differs substantially from an explosive-price screen, consistent with its design of adjusting for common variance movements. The 1990s calm-start rate is higher, so the contrast is specific to these cohorts.

8.3 Homebuilders: a coherent early sector signal

The four homebuilders provide the clearest daily case study. The monitor flags D.R. Horton and Lennar in December 2003, then Pulte and Toll Brothers in January 2004. Their respective price peaks occur in July 2005. All four alarms precede the peaks by 377–411 trading sessions, a remarkably similar lead across the sector. GSADF flags only Toll Brothers, 112 sessions before its peak.

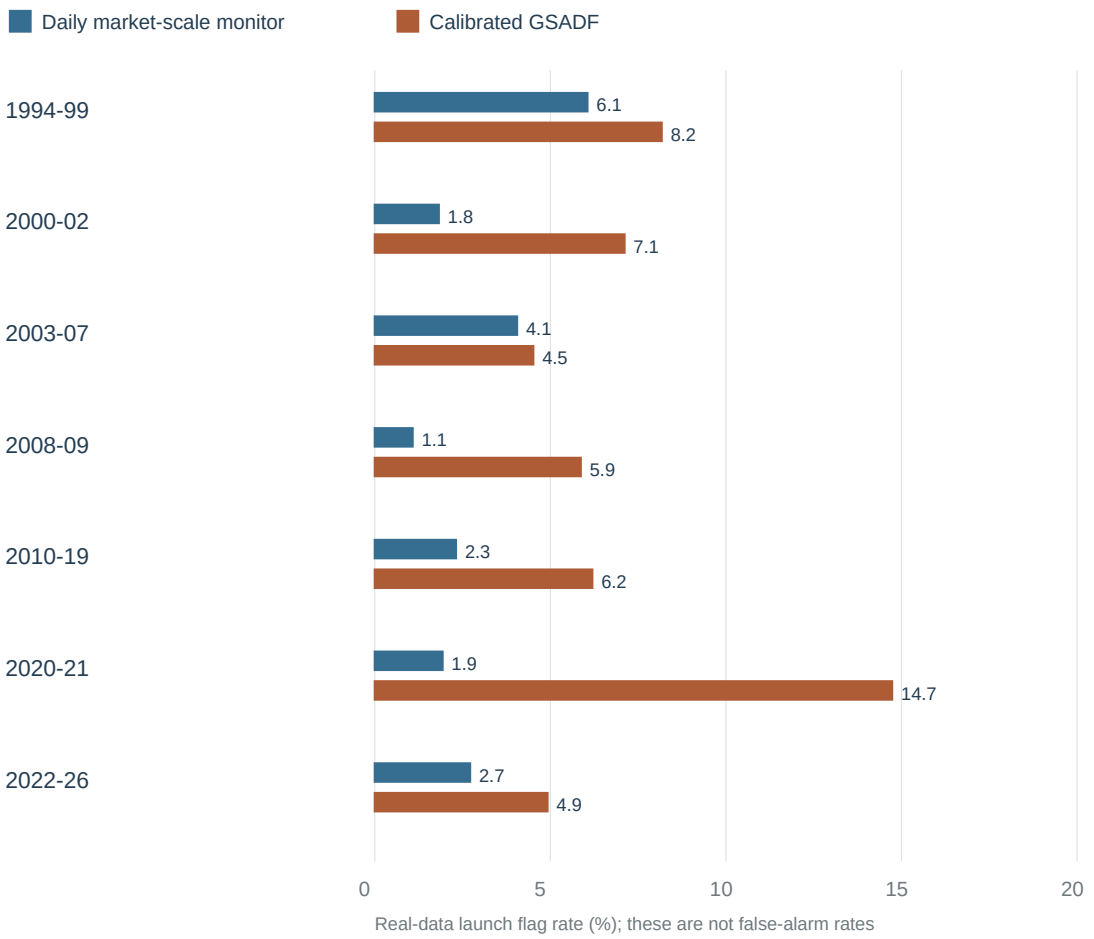


Figure 2: Daily monitor and calibrated GSADF rates by launch cohort. Both procedures use the same universe and earlier reference records. Source: experiment 056.

Name	First daily alarm	Price peak	Daily lead	GSADF lead
D.R. Horton	2003-12-10	2005-07-20	404	—
Lennar	2003-12-09	2005-07-28	411	—
Pulte	2004-01-09	2005-07-28	390	—
Toll Brothers	2004-01-21	2005-07-20	377	112

Table 6: Homebuilder alarms in the registered 2003–08 records. Leads are trading sessions before each record's price maximum; a dash denotes no alarm. Source: experiment 056.

This is evidence of a shared developing price–variance pattern well before the sector's peak. For monitoring, identifying that development is a useful distinct objective from predicting the date of reversal. The table records first crossings; it does not measure the subsequent persistence or intensification of the signal.

Across the wider collection, the daily monitor alarms before the peak on 15 of 30 supported consensus-positive records, against 14 for GSADF; on the collapsed-name collection, the counts are 4 of 9 against 3. On 18 recent narrative records, they are 7 against 10. The median pre-peak consensus lead is 377 sessions versus 142 for GSADF. Results differ within financials: BAC, JPM, AIG, and FNMA trigger after their peaks, while Lehman and Countrywide trigger 107 and 833 sessions before theirs.

Two comparison conventions matter. Across consensus-negative records, 4.8% of launches alarm and 35 of 91 records contain at least one alarm; monthly relaunching creates more opportunities than a single launch. GSADF also uses prior-inclusive history: 315 of its 650 alarmed episode launches cross on the first scored day. These are comparisons of the specified operational procedures.

Research record: experiment 056, including its denominator and timing readings.

8.4 How this differs from the repaired own-state score

The earlier repaired daily score used a stock-specific volatility state. On 10,000 universe windows, its empirical rate was 4.9%, but yearly rates ranged from about 1.4% to 16.2%. It flagged only 3 of the 33 requested consensus positives, while flagging 20 of 103 calm controls; three non-US positive records lacked complete histories on the imposed US calendar. Its consensus ranking had an AUC of 0.34, with an interval containing 0.5. That repair had little episode discrimination under its specified design.

On common windows, the market-scale statistic has a rank correlation of about 0.34 with that own-state score. Their distinct outputs show that changing the background-state treatment materially changes the measurement.

Research record: experiments 051 and 056.

8.5 What the earlier alarms contain

An audit of 185 flags from the earlier historical comparisons found that 156 survived a declared jump filter, including 137 of 153 explosive-root flags. In daily records, rising volatility through the window was the more distinctive feature: 69% of flags exceeded the controls' 95th percentile of the second-half/first-half volatility ratio. Removing jumps increased the daily calm GSADF rate from 19.4% to 25.2%. This helps explain why background variance and empirical calibration matter more than simply deleting extreme returns in these comparisons.

Research record: experiment 041; the audit concerns the earlier scores, not a new jump-filter evaluation of the daily market-scale monitor.

8.6 Implied volatility as a different background scale

A separate construction used a lagged VIX or VXN value to scale a declared Gaussian null. In simulation it had daily power against level dependence not already reflected in that scale; dependence tracked by the supplied scale was absorbed. In real SPX and NDX windows the registered scores were silent, including the named historical episodes, apart from a few crossings in a narrower sensitivity band. Some variance ratios fell outside the declared band, and VIX before 22 September 2003 is a retrospective backcast, making the earlier SPX windows reconstructed-scale analyses. A separate stricter as-of design was withdrawn before scoring. The daily market-scale method above uses a different background adjustment.

Research record: claim 042; experiments 043 and 047.

9 What five-minute data add

The intraday evidence first examines the pattern, then evaluates the specified monitor. These are different exercises: a coefficient estimated on a whole retrospectively chosen episode is not an alarm available at its beginning.

9.1 Within-session variance behavior

On the descriptive collection, the median within-day variance exponent is 2.35 for four meme-stock records and 1.48 for 20 narrative-name records. Four growth-name records have median −3.59; calm Dow members have median −5.94. Most index estimates hit the lower search boundary of −20, which is a numerical boundary, not an estimate beyond that boundary. The meme records show positive fitted effects, while the growth collection and some prominent narrative names show the usual negative direction.

The pattern is less pronounced when one isolates hindsight-defined run-up segments, and the estimates also respond to return–volatility correlation in simulation. The empirical result is a difference in measured price–variance behavior across these groups, encompassing both local-level and correlated-volatility interpretations discussed in Section 4.1.

Research record: experiments 052–053; model-based curvature errors for the within-day fits are distinct from empirical calibration.

9.2 A retrospective replay of the monitor

The specified monitor is calibrated on 499 calm-start references from 2017–18, all completed before targets begin in February 2019. It then replays each 21-session launch with an assumed one-minute availability delay. GSADF is evaluated at each daily close using the 252 prior closes and the monitored sessions so far, with its own cutoff fitted to the same reference IDs.

Records	Launches	Intraday monitor	GSADF
All run-up launches	449	19.6	40.5
Meme stocks, 2020–21	52	46.2	67.3
Growth names	164	9.1	25.6
Narrative names, 2023–26	233	21.0	45.1
Same-date calm members	805	4.2	0.0
Same-date SPY and DIA	322	0.6	0.0

Table 7: Launches with an alarm within the 21-session horizon (%). The run-up denominator retains training failures. The control rows reuse name-launches across episode windows and are descriptive. Source: experiment 054.

The intraday monitor alarms on 46.2% of meme-stock launches. Across all run-up records the rate is 19.6%. GSADF alarms more often on these episodes and on none of the selected calm controls. Of 29 run-up records, the monitor alarms strictly before the peak on 14, GSADF on 20, with one additional GSADF alarm on the peak date. The monitor's median pre-peak lead is 142 sessions, compared with 57.5 for GSADF. Seven monitor alarms precede the defined run-up onset; only seven fall between onset and peak.

There are important availability and timing differences. Of 449 run-up launches, 109 fail before fitting because fewer than 200 prior sessions have at least 70 bars. Seven records never obtain a successful launch. The failures remain in the rates. GSADF uses the available daily history, and 163 of its 182 alarmed positive

launches cross on the first monitored session. Thus the table compares procedures that can be deployed under the declared rules, with GSADF benefiting from more daily history at launch.

The calm-control table has 1,127 weighted entries but only 610 distinct name-launches. The monitor alarms on 30 of those distinct launches, 4.9%; the weighting in the displayed table yields different subgroup rates. Inference must account for repeated names and dates. In an unselected 100-name draw, the monitor alarms on 5.0% of 8,600 launches, versus 5.4% for GSADF, with its own annual rate varying from 2.3% to 9.8%. These frequencies describe the broad-draw population under the fixed reference calibration.

The direct daily/intraday comparison also favors the faster observations for early recognition. Among the 29 common run-up records, both monitors alarm somewhere in 14, with the intraday alarm earlier in 13 of those. Five records have only a daily alarm, six only an intraday alarm, and four neither. Counting strictly pre-peak alarms, the totals are 13 daily and 14 intraday. These are the registered record definitions; three names have different peak dates in the two source experiments.

Research record: experiment 056, comparison with experiment 054.

9.3 Incremental out-of-sample drawdown information

The intraday score supplies the first positive incremental out-of-sample drawdown result in this sequence of lab scores. Adding it to GSADF raises predictive R^2 from 1.05% to 1.39% at 63 sessions and from 1.49% to 1.74% at 126 sessions. Thus the intraday statistic contains information that the explosive-root score alone leaves out of these fitted forecasts.

Regressions use within-launch-date ranks, an intercept, and either or both scores. Outcomes start after the completed scoring window. Coefficients are trained on records whose outcomes end before 2023 and evaluated on launches from 2023 onward.

Forward sessions	Intraday alone	GSADF alone	Combined
63	0.39	1.05	1.39
126	0.31	1.49	1.74

Table 8: Out-of-sample R^2 (%) for maximum drawdown, relative to the training mean. The rows concern completed-window scores, not first alarms. Source: experiment 054.

The gains of 0.34 and 0.25 percentage points are point estimates; paired uncertainty for the improvements was not computed. These score-only regressions also leave open how much information survives controls for realized volatility and momentum. This is the scope of the forecasting finding, which is supplementary to the measurement and alarm results. Neither score demonstrates return predictability in this exercise. The daily market-scale score's drawdown R^2 is much smaller, approximately 0.03–0.16%, with no demonstrated useful incremental gain over GSADF.

10 Reading and using an alarm

The proposed signals answer a conditional measurement question. An alarm means that the fitted positive-level alternative accumulated unusually large evidence relative to the declared comparison model and reference records. It is useful to inspect the estimated level pattern, background scale, data support, and the behavior of alternative calibrations together. A large raw likelihood contrast with an unusual reference history is a different finding from a stable contrast across several predeclared reference choices.

The daily and intraday results suggest complementary descriptions. The daily market-scale construction produces relatively sparse alarms and has demonstrable model-based power when the market proxy tracks the nuisance. The intraday construction detects within-session patterns in some speculative episodes and adds out-of-sample drawdown information. GSADF remains the stronger general episode alarm in the intraday comparison. The new scores contribute different measurements: selective daily patterns and an intraday component that improves the combined forecast.

For ongoing monitoring, two further results from the research matter. First, repeatedly relaunching a monitor changes the error event. The exact-model framework can fund declared or adaptive restarts from one budget; it cannot discard unfavorable evidence and restart for free. Second, dependence across names does not invalidate e-BH when genuine e-values are supplied, but it does not make an empirical score an e-value. Reviewer simulations of pooled forecasts found no improvement over per-name alternatives. Experiment 029 likewise found essentially no detection gain from selecting a cross-sectional clock within its specified model; outside that model, selection could damage false-discovery control.

The construction of an exact filter for rounded prices is another useful methodological result. It treats the hidden efficient price jointly across the record instead of paying an independent worst-case penalty at each observation. That result belongs to an exact observation model. Applying it to recorded trade closes with different noise and rounding mechanisms requires a new model check, not merely a finer sampling interval.

Part III

Rebalancing and Traded Funds

How finite trading, costs, and liquidation change the expectation shortfall.

11 Finite rebalancing changes the classification

The continuous-exposure result is not automatically a theorem about an exchange-traded fund that resets its exposure once a day. Between reset dates, the elementary contract holds fixed shares and a cash position. This distinction changes the martingale classification even before fees, financing, or tracking error are introduced.

Let X be a continuous nonnegative Q-local martingale with $X_0 = 1$. Over one holding interval, normalized gross wealth at exposure π is

$$G_t^\pi = 1 - \pi + \pi X_t. \tag{18}$$

Writing $\delta_T = 1 - \mathbb{E}_Q[X_T]$, linearity gives

$$1 - \mathbb{E}_Q[G_T^\pi] = \pi \delta_T. \tag{19}$$

For $\pi < 0$ this is a negative signed gap when the underlying is strict. It is not a negative bubble in a nonnegative asset: the affine contract can itself become negative. Limited liability needs an additional rule.

11.1 Finite grids with exposure between zero and one

For a finite deterministic grid $0 = t_0 < \cdots < t_N = T$, define

$$W_t = W_{t_i}\left(1 - \pi + \pi \frac{X_t}{X_{t_i}}\right), \qquad t_i \le t \le t_{i+1}, \quad X_{t_i} > 0. \tag{20}$$

When $X_{t_i} = 0$, keep the remaining wealth in cash. For $0 < \pi \le 1$, the factors and the wealth remain nonnegative.

Proposition 5 (Finite-grid inheritance)**.** *For $0 < \pi \le 1$ and every finite grid, W is a nonnegative continuous local martingale and is a true martingale on $[0, T]$ exactly when X is. If X is strict on that horizon, so is W.*

The mechanism is the accumulated conditional expectation loss of each holding. Define $A_i = X_{t_i} - \mathbb{E}_Q[X_{t_{i+1}} \mid \mathcal{F}_{t_i}] \ge 0$ and $h_i = \pi W_{t_i}/X_{t_i}$ where the denominator is positive, zero otherwise. Then

$$1 - \mathbb{E}_Q[W_T] = \sum_{i=0}^{N-1} \mathbb{E}_Q[h_i A_i]. \tag{21}$$

Underlying, $\pi = 1/4$	Asset	Every finite grid	Continuous wealth
Absorbed Brownian, $\sigma = 1$	Martingale	Martingale	Strict
Reciprocal-Bessel type, $\sigma = x^2$	Strict	Strict	Martingale

Table 9: The discrete/continuous reversal under the stated wealth and post-default conventions. These are pricing-measure classifications.

Each summand is nonnegative. If X is a martingale, every A_i vanishes. Conversely, induction gives a lower bound on the holding where A_i can be nonzero, and $1 - \mathbb{E}_Q[W_T] \geq \pi^N \delta_T$. This proves inheritance. The bound can be extremely small and is generally useful for classification rather than for monetary estimation.

The two examples in Part I now become a comparison of trading rules.

At default, a finite-grid fund preserves the cash fraction left from the last rebalance. The continuous model keeps replenishing the risky fraction as the price approaches zero. In the absorbed Brownian example, every finite-grid terminal mean is one while the continuous mean is smaller, so the terminal values cannot converge in L^1. This does not by itself rule out weaker modes of convergence through absorption.

Research record: claims 004, 005, and 001. The stochastic inputs are finite trading, bounded stopping, and the nonnegative-local-martingale supermartingale property.

11.2 Refinement and the size of the discrepancy

Away from zero, standard stability of stochastic differential equations gives convergence of the grid recursion to the stochastic exponential as the mesh shrinks [28]. Localizing to $X \in (1/m, m)$ gives uniform convergence in probability on a finite horizon. If X stays strictly positive throughout the horizon, the stop can be removed. The stability theorem supplies the path-convergence step.

Path convergence does not alone ensure convergence of expectations of unbounded payoffs. Bounded Lipschitz functionals, including puts on nonnegative wealth, have converging expectations. Calls require additional tail control. For $0 \leq \pi \leq 1$ and $Q(X_t > 0 \text{ for all } t \in [0, T]) = 1$, a mean-one continuous limit supplies that control automatically:

$$\mathbb{E}_Q[V_T^\pi] = 1 \quad \Longrightarrow \quad \mathbb{E}_Q|W_T^{(n)} - V_T^\pi| \longrightarrow 0. \tag{22}$$

Fatou's lemma and the bound $\mathbb{E}_Q[W_T^{(n)}] \leq 1$ force the means to converge to one; the nonnegative shortfall then gives L^1 convergence. All Lipschitz payoff expectation prices, calls included, converge.

For $\sigma = x^2$ and $\pi = 1/4$, every finite-grid fund is strict yet its defect tends to zero. Binary classification can therefore differ at every finite frequency while the valuation discrepancy disappears. When the continuous limit is strict, the existing result gives only a one-sided bound on the limiting defect, not full convergence. A numerical study at $\pi = 3/4$ finds means approximately 0.786, 0.819, and 0.833 for 4, 64, and 1,008 intervals, approaching the continuous value 0.8465. That pattern is evidence in one model; it cannot exclude probability mass in paths too rare for the numerical calculation to resolve.

Research record: claims 007 and 009; experiment 012.

12 Leverage, inverse exposure, and liquidation

For $\pi > 1$, a sufficiently large decline makes (18) negative. For $\pi < 0$, a sufficiently large rise does the same. A model of a nonnegative fund must say what happens at that point. Resetting the exposure at the next date is not enough to specify limited liability between dates.

12.1 Inverse funds under an explicit floor rule

Let W now be the signed finite-grid recursion at $\pi < 0$. Fix $0 \le a < 1$ and liquidate into cash at the first continuous crossing below the floor a, capped at T. By continuity, wealth at an actual crossing is a. Denote stopped wealth by H^a. On a grid with N intervals,

$$a \le H_t^a \le (1-\pi)^N. \tag{23}$$

A bounded local martingale is a uniformly integrable martingale. Thus this finite-grid, continuously liquidated inverse fund has no defect, even when its underlying is strict.

For example, at $\pi = -1$ on one interval, liquidating when X first exceeds $3/2$ preserves wealth between $1/2$ and 2. The underlying $\mathrm{d}X = X^2\mathrm{d}W$ can be strict while the liquidated inverse contract is a martingale. This is an implication of the contract rule, not evidence that inverse exposure removes every economically relevant risk.

12.2 Leveraged-long funds

For $\pi > 1$, impose the same floor rule and write $d_H = 1 - \mathbb{E}_Q[H_T^a]$. The lab establishes

$$a \le H_t^a \le a + \pi^N X_t, \qquad 0 \le d_H \le 1 - a, \qquad d_H \le \pi^N \delta_T. \tag{24}$$

Hence a true-martingale underlying gives a true-martingale liquidated fund. A positive fund defect implies a positive underlying defect, but the converse is not established. Since π^N grows rapidly, the quantitative lower bound $\delta_T \ge d_H/\pi^N$ is weak at daily rebalancing frequencies.

In the registered simulations, leveraged-long funds over two strict underlyings retain defects of roughly 0.40–0.96 at daily rebalancing for exposures 1.5–3 with a zero floor, and 0.36–0.50 with a half-wealth floor. Inverse funds over the strict underlyings are consistent with zero. High-leverage geometric-Brownian controls are less stable: some inverse cells are unresolved and the long controls at exposure three are fragile individually. The large long-fund effects are results for these chosen models and floor rules.

Research record: claims 011 and 018; experiments 017–018.

12.3 Costs are not a bubble

If terminal net wealth is $Y_T = W_T - c_T + e_T$, where costs have expected value d_T under Q and $\mathbb{E}_Q|e_T| \le \varepsilon$, then

$$\left|(1 - \mathbb{E}_Q[Y_T]) - (1 - \mathbb{E}_Q[W_T]) - d_T\right| \le \varepsilon. \tag{25}$$

Expense, borrowing, and financing terms can be incorporated into the one-interval return accounting, following the leveraged-fund framework of Avellaneda and Zhang [1]. A deterministic fee can lower expected net wealth without any strict-local-martingale mechanism. An expected tracking bound under P does not automatically provide the required bound under Q.

The measured ETF data put a scale on these adjustments. Across eight leveraged and inverse funds on QQQ and SPY, leveraged-long funds lose an additional 0.6–1.7% per year beyond the expense ratio, and

inverse funds 0.1–1.1%, measured against the total-return underlying. Reported standard errors are 0.1–0.4 percentage points per year. Using a price index instead would misread distributions as a cost. This is useful return accounting: it separates ordinary financing and tracking effects from the expectation shortfall studied in the theoretical model.

Research record: claim 004; experiment 007.

13 What discrete monitoring misses

Liquidation is path dependent. A simulated path can cross a boundary and recover between observations. A program that looks only at observation times can keep a fund alive after the continuously monitored contract has terminated.

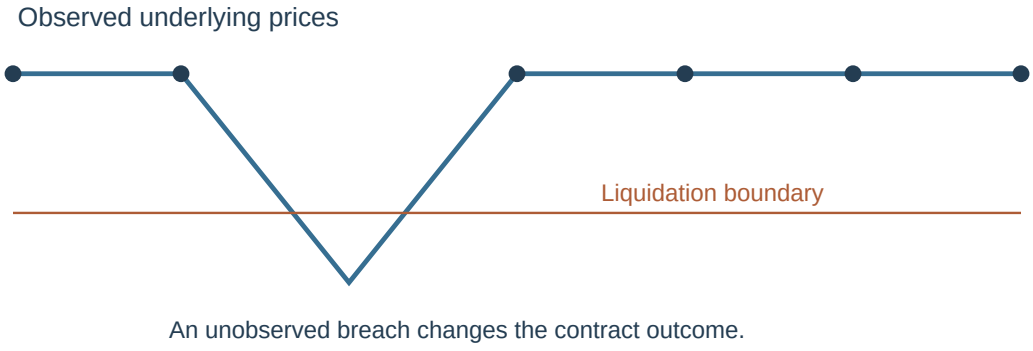


Figure 3: A boundary breach can be invisible at every observation time. This is an explanatory schematic, not a simulated or historical record.

For monitoring times F containing all rebalancing dates and T, define a numerical quantity Z_F: if a sampled value of the signed recursion is below a, replace terminal wealth by a; otherwise keep W_T. This clipping is an artificial cash adjustment, not an executable trading strategy. It has the useful pathwise property

$$a \le H_T^a \le Z_F, \qquad 1 - \mathbb{E}_Q[Z_F] \le d_H. \tag{26}$$

Adding monitoring times can only decrease Z_F. On a fixed rebalancing grid, dense monitoring gives L^1 convergence to continuously liquidated wealth under the stated models.

There is no uniform error rate based on the mesh alone over all continuous paths. For any finite set of monitoring times, insert an unobserved downward excursion in a gap. With one interval, $\pi = 2$, and $a = 1/2$, a path observed at $X = 1$ but dipping to $X = 1/2$ in between has $Z_F = 1$ and liquidated terminal wealth $H_T^a = 1/2$. Continuity alone places no bound on how narrow the excursion can be.

This result gives a constructive numerical lesson. The particular clipped quantity bounds the defect from below in expectation; freezing wealth at the first sampled overshoot generally does not. A positive Monte Carlo estimate still has sampling and discretization error, so it is evidence for a positive lower bound, not a proof of its sign. The fund simulations use this distinction to check the direction of monitoring bias.

Research record: claim 020; experiment 018.

Part IV

Options and Constituent Information

What additional contracts reveal, and which valuation assumptions make that possible.

14 Option prices do not have a unique bubble interpretation

The option results explain how additional contracts can constrain a valuation amount. The practical difficulty appears in the observed quotes: on the sampled end-of-day SPY chains, the largest-strike calls at one to three months sit at the one-cent floor on most dates, at median strikes 1.07–1.41 times spot. The available far wing is therefore measured only to the tick. Reading such quotes as a bubble bound can produce a very small number largely determined by the pricing convention and quote resolution.

Research record: experiment 014.

The central issue is whether the observed call represents an expectation price, a parity-restoring price, or an exercise value. Assuming that a call carries the defect and then reading a tiny quote as evidence of its absence makes the conclusion depend on the premise being assessed. The results below are consequently valuation identities and bounds under named conventions; the research did not obtain a usable market bubble test from these quotes. The algebra is nevertheless informative about which observations would constrain a defect and which can leave it hidden.

Normalize spot to one, take zero rates and dividends, and fix a common pricing law with $m = \mathbb{E}_Q[X_T]$ and $\delta = 1 - m$. European expectation prices satisfy

$$c(K) = \mathbb{E}_Q[(X_T - K)^+], \quad p(K) = \mathbb{E}_Q[(K - X_T)^+], \qquad c(K) - p(K) = m - K. \tag{27}$$

Thus $1 - K - c + p = \delta$. Under another declared rule, leave the put at p and set the call to $C = c + \delta$. Then $1 - K - C + p = 0$ even when $\delta > 0$. This second rule restores ordinary parity.

The distinction builds on the strict-local-martingale option literature. Cox and Hobson explain how parity and exercise values depend on the admissible trading strategies [5]. Carr, Fisher, and Ruf obtain parity restoration from their joint-superreplication setting [4]; it is not a universal conclusion that market calls follow that rule. The lab uses the two rules as explicit alternatives and works out their implications for inference. Calling both simply "risk-neutral option prices" would hide the assumption that determines the answer.

14.1 Exercise value and the parity residual

For the corresponding continuously exercisable American call, the lab's localization argument gives $A_Q \geq c + \delta$. Write $A_Q = c + \delta + u_A$, where $u_A \geq 0$ denotes any excess over that lower bound. With a European expectation put in every row:

Contract metadata distinguish American from European exercise even if their prices coincide. Prices alone need not distinguish valuation mechanisms: on the equality face $u_A = 0$, the parity-restored European value and the American value agree. Replacing the put by an American put adds its own exercise

Call convention	Call value	$1 - K - C + p$	Reading
European expectation	c	δ	Residual equals the defect
European parity-restoring	$c + \delta$	0	A defect can be hidden by parity
American exercise	$c + \delta + u_A$	$-u_A$	Residual concerns exercise excess

Table 10: Residuals under declared valuation rules. The table is not a classification algorithm for market quotes.

premium and changes the table. Neither a zero residual nor an American premium alone identifies a defect without the rest of the model.

With simultaneous intervals for the relevant operator prices, one can propagate errors through the residual and test the zero-residual restriction. If the total call-plus-put error is bounded by h, a positive residual exceeding $2h$ in the expectation model is sufficient for rejection by a conservative lower-bound rule. That is a conditional separation statement. A bid–ask spread by itself does not establish the required coverage of a specified pricing operator.

Research record: claims 014 and 017; the D31 option-observables design. This design contains no real-data option test.

15 Which option observations are informative?

15.1 The lowest strike matters

Suppose only expectation-priced calls are observed at finitely many strikes $K_1 < \cdots < K_r$, all at or above normalized spot one. If their prices are compatible with a nonnegative terminal law of mean at most one, then they are also compatible with a terminal law of mean exactly one. They cannot exclude zero defect.

The construction is simple. Starting from a compatible random variable X with mean $m < 1$, compare it with $Y = \max(X, K_1)$. Every observed call payoff is unchanged because the added mass lies below its strike. But $\mathbb{E}[Y] = K_1 + c(K_1) \geq 1$. Mixing the two terminal laws raises the mean to one while preserving the entire observed call panel.

This is a statement about terminal laws, not a construction of a diffusion for every feasible law. It still makes a useful observation-design point: adding ever-higher call strikes does not automatically identify the mean missing below the lowest strike. An in-the-money call can exclude mean one in this model. A matched underlying call and put identify the mean directly through (27), conditional on expectation pricing.

The lab also constructs computable outer intervals for the defect using call monotonicity, convex secant slopes, and quote intervals. Those inequalities alone are not always a sharp characterization of attainable terminal laws. Inconsistent observations must be reported as model incompatibility, not automatically as positive bubble evidence.

Research record: claim 021.

15.2 Do options on leveraged funds add information?

For a normalized fund, the same expectation-pricing identity yields its defect from a call–put pair. Finite-grid inheritance then links a positive fund defect to the underlying defect. For exposures $0 < \pi \leq 1$, the quantitative link uses the factor π^N; for liquidated long leverage it uses π^N in the opposite inequality. Both become weak for monetary estimation at realistic numbers of resets.

For example, 0.75^{252} is about 3×10^{-32}. An upper bound on the fund defect divided by this factor is normally uninformative about the underlying's defect. On the leveraged side, a positive fund defect divided by π^N supplies a negligible lower bound. Zero exclusion can remain mathematically meaningful even when amount estimation is weak, but its practical power then depends on resolving a small residual amid costs, tracking, and quote uncertainty.

If an exact expectation-priced underlying call–put pair is already available, it identifies the defect without fund options. Adding fund options cannot improve on exact identification. With only underlying out-of-the-money calls, a positive fund residual can add a restriction, but no empirical gain specific to leverage has been demonstrated. The inverse liquidation model in Part III has zero fund defect even over a strict underlying, so it cannot supply the same zero-exclusion logic.

Research record: claims 014, 018, and 021; no real-data fund-option detection result is claimed.

15.3 Continuous exercise creates another limiting distinction

Let $G_t = \bar{S}_t + D_t$ be a finite continuous nonnegative Q-local martingale, where $\bar{S}_t = d_t S_t \geq 0$ and d_t is a positive deterministic continuous discount factor, bounded on $[0, T]$. The cumulative discounted distributions D are adapted, nonnegative, nondecreasing and càdlàg, with $D_0 = 0$. A distribution jump may be offset by a price drop while the gain process remains continuous. Write $b_G(T) = G_0 - \mathbb{E}_Q[G_T]$, and define

$$C_E(K,T) = \mathbb{E}_Q[(\bar{S}_T - d_T K)^+], \qquad A_Q(K,T) = \sup_{\tau \leq T} \mathbb{E}_Q[(\bar{S}_\tau - d_\tau K)^+],$$

where the supremum includes every stopping time in $[0, T]$. For this continuously exercisable American call, localization at high gain levels and bounded convergence of the capped payoff component give

$$A_Q(K,T) \geq b_G(T) + C_E(K,T) \geq b_G(T). \tag{28}$$

The relation between American exercise and bubbles has established antecedents [3]; it is not presented as a new general pricing principle. The lab's treatment includes distributions and the transfer to fixed-basket bounds.

The contrast with finite exercise is especially revealing. European expectation calls tend to zero as $K \to \infty$. The same is true when exercise is restricted to any fixed finite set of dates, because the exercise payoff is bounded by a finite sum of integrable call payoffs. Such prices therefore cannot uniformly bound a positive defect from above as the strike increases. Continuous exercise permits stopping at rare high levels in a way that a fixed finite exercise schedule does not. The order of these limiting operations matters.

A tiny quoted call is consequently not an assumption-free certificate of a tiny bubble. To use (28), the quote must provide a justified upper bound on the specified continuous-exercise value. The model of exercise and its relation to observed market prices are substantive premises.

16 From an index to its constituents

For fixed positive holdings n_i and a common pricing measure, let $I_t = \sum_i n_i S_{i,t}$, $\omega_i = n_i S_{i,0}/I_0$, and $b_i(T) = S_{i,0} - \mathbb{E}_Q[S_{i,T}]$. Linearity gives

$$b_I(T) = \sum_i n_i b_i(T), \qquad 0 \leq n_i b_i(T) \leq b_I(T). \tag{29}$$

This is a consequence of the definition, not a new diversification principle. It concerns fixed holdings under one measure; a rebalanced index needs its trading rule specified separately.

16.1 Ordinary weights can hide all of the effect

Suppose two names are proportional strict prices, with equal initial weights. Their ordinary cap weights remain exactly one-half. Both can have positive defects while neither weight declines. Thus a proposed attribution based solely on expected losses of ordinary weights cannot work in general.

The corrected representation uses the index share probability. Its defining identity carries an explicit survival event up to a lifetime τ^I [16]. Let $\mu_i(t) = n_i S_{i,t}/I_t$ and $\mu_i^{\dagger}(t) = \mu_i(t)\mathbf{1}_{\{t<\tau^I\}}$. Then

$$\omega_i - \mathbb{E}_{\widetilde{Q}_I}[\mu_i^{\dagger}(T)] = \frac{n_i b_i(T)}{I_0}, \qquad \sum_i \mu_i^{\dagger}(T) = \mathbf{1}_{\{T<\tau^I\}}. \tag{30}$$

These survival-adjusted, or killed, weights need not sum to one after the lifetime. Ordinary weights do, which is why they cannot all be supermartingales with a positive total expected loss under a single probability. In the proportional example, all the loss is carried by the survival factor.

The associated index hazard is also a survival-weighted mixture. If $p_i(t) = \mathbb{E}_Q[S_{i,t}]/S_{i,0}$ are absolutely continuous with $h_i = -p_i'/p_i$, then

$$p_I = \sum_i \omega_i p_i, \qquad h_I = \sum_i \frac{\omega_i p_i}{p_I} h_i. \tag{31}$$

The weights are not simply today's or the initial cap weights. These are auxiliary-measure lifetimes and hazards, not directly observed physical crash probabilities.

Research record: claim 010.

16.2 A conditional recovery formula

Let ν_T be the pricing distribution of the terminal index. The recovery result takes this distribution as a supplied population input. Let $m_i(x) = \mathbb{E}_P[\mu_i(T) \mid I_T = x]$. If the terminal change of measure has the form $\mathrm{d}Q/\mathrm{d}P = k_T(I_T)$, then conditioning on the terminal index preserves the conditional mean weights. Hence

$$\frac{n_i b_i(T)}{I_0} = \omega_i - \frac{1}{I_0}\int x m_i(x)\nu_T(\mathrm{d}x). \tag{32}$$

The weaker condition that the conditional mean weights agree under P and Q already suffices. The density condition is one way to imply it, and is much stronger than loosely saying that the pricing kernel "depends on the market."

The formula is an identification result under that restriction, not an estimator from one historical stock panel and today's finite option chain. The required conditional means and pricing distribution are population objects. The lab's estimator assumes independent physical terminal panels and an independent supplied pricing sample; it does not construct those inputs from overlapping historical windows.

There is a necessary compatibility check: $0 \le \int x m_i(x)\nu_T(\mathrm{d}x) \le n_i S_{i,0}$ for every name. The estimated contributions summing to the index contribution is not a check, because the identity follows automatically when the conditional weights sum to one. This distinction is useful whenever an attribution method appears to validate itself by matching a total.

Research record: claims 016 and 044.

16.3 What the synthetic estimation says

The recovery estimator preserves the total on every sample and is consistent under the stated sampling and approximation assumptions. A histogram-based necessary-compatibility test has a conditional finite-sample error bound with a supplied smoothness constant. In controlled terminal-law examples, finite-sample tests detect large violations; the fixed partitions used there do not implement an asymptotic power construction.

In the more process-like basket example, however, the bubble contribution is about 0.0041 while the fixed-partition estimator bias is about 0.0035. Increasing the sample size does not remove fixed-range and fixed-partition bias. No compatibility rejection was observed in the basket cases tested. A numerical smoothness estimate gives convenient intervals but lacks a rigorous bound; the available rigorous bound is too large to make those intervals useful. The population identity is stronger than this estimator's demonstrated performance.

Research record: experiment 046.

16.4 What a basket budget can and cannot identify

Suppose an option on a tracking fund supplies a justified upper bound U on its American exercise value, and η bounds the expected tracking discrepancy under Q relative to a frozen basket. With the appropriate scaling a, the total defect is bounded by $B_{\mathrm{cap}} = aU + \eta$. Then

$$\frac{b_i(T)}{S_{i,0}} \le \frac{B_{\mathrm{cap}}}{\omega_i I_0}. \tag{33}$$

The same absolute budget can give an informative relative bound for a large constituent and a vacuous one for a tiny constituent. Tracking under the physical measure does not supply η, and fees and basket reconstitution can matter more than a small quoted wing call.

An upper budget alone never forces any particular name to have a positive defect. Combined with individual intervals and a positive lower bound on the total, the lab derives exact coordinate bounds for the resulting box intersected by a weighted-sum constraint. That finite-dimensional calculation is useful bookkeeping, but it cannot create the missing monetary lower bound from a binary price-based signal. A small positive upper bound is a tolerance certificate, not proof of exact absence.

Research record: claims 017 and 019.

17 What the four parts establish

The research gives a connected account of three practical objects: a price–variance signal, the wealth produced by a trading rule, and an expectation shortfall under a pricing law. The daily statistic detects specified level alternatives substantially more often than calibrated GSADF and identifies a coherent early pattern across the four homebuilders. The intraday score adds drawdown information to the explosive-root benchmark. Calibration determines the reference against which either pattern is unusual.

The theory explains why those measurements and a valuation defect are related but distinct. Rapidly growing local volatility and the positive shock alignment in Sin's model provide two mechanisms. The exposure classification shows how rebalancing can change strictness, and the finite-grid and liquidation results explain why actual contract rules matter. A bounded liquidated inverse fund and an unbounded leveraged-long fund can have different expectation properties over the same underlying.

Options and constituent data then put structure on the valuation question. A matched expectation-priced call and put determine a terminal mean; a high-strike call panel alone can leave it unidentified. A fixed basket adds constituent defects, while attribution needs more information than ordinary weights. These results give the practitioner useful distinctions: which pattern a monitor measures, which exposure a contract creates, and which assumptions make a monetary bound informative.

A Notation and algorithm specifications

A.1 Common notation

Symbol	Meaning
P, Q	Physical observation law and a specified pricing measure.
$b_S(T)$	Monetary defect $S_0 - \mathbb{E}_Q[S_T]$.
δ_T	Defect of a price normalized to start at one.
$\sigma(x), p$	Price diffusion coefficient; CEV price exponent in $\sigma(x) = \kappa x^p$.
I_0, I_∞	Lower and upper integrals of $x/\sigma^2(x)$.
π	Portfolio exposure, not a significance level.
W, V^π, H^a	Finite-grid wealth, continuous wealth, and floor-liquidated wealth.
β	Full variance/price exponent; $\beta = 2(p-1)$ in the CEV motivation.
A_t, v_t	Background variance and variance including the level term.
μ, ν	Return location and Student-t degrees of freedom.
M_t, m_t	Lagged market squared-return scale and its training-normalized value.
h_d, c_j	Intraday background state for session d and diurnal slot clock.
α	Per-record calibration target.
$n_i, \omega_i, \mu_i(t)$	Fixed basket holdings, initial cap weights, and current cap weights.
ν_T, η_T	Index pricing distribution and its size-biased subprobability measure.

Earlier papers and code use the letter β for several different objects, including the CEV price exponent, the exponent of standard deviation, and the bubble amount. They are not identified with one another here. Monetary defects use b, CEV price elasticity uses p, and the statistical variance exponent uses β. The common D35 fitter used by experiments 050–051 and the D36 monitor in experiment 054 store $\beta/2$ in the present notation. Experiments 049, 052–053, and D37 use the full variance exponent. Historical power figures quoted without an exponent should not be used to infer a different scaling convention.

A.2 Student-t density and the numerical profile

For variance parameter v and $\nu > 2$, the variance-one Student-t log density is

$$\log f(r; \mu, v, \nu) = \log\Gamma\left(\frac{\nu+1}{2}\right) - \log\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\log\big(\pi(\nu-2)v\big) - \frac{\nu+1}{2}\log\left(1 + \frac{(r-\mu)^2}{(\nu-2)v}\right). \tag{34}$$

With $v_t = A_t e^{\beta x_{t-1}}$, write $z_t = (r_t - \mu)^2/((\nu-2)A_t e^{\beta x_{t-1}})$. The derivative of a log-density contribution is

$$\frac{x_{t-1}}{2}\left(-1 + (\nu+1)\frac{z_t}{1+z_t}\right),$$

and the second derivative is nonpositive. This concavity motivates the half-line profile search. A boundary derivative is checked first; otherwise the implementation doubles the search magnitude up to 2^{30},

then bisects a derivative bracket to width 10^{-10}, with at most 200 bisections and 400 objective/derivative evaluations. These are floating-point stopping rules, not outward error enclosures.

The profile may be flat, unbounded, or unresolved. Each case has a separate status. A failed required profile produces the failure score -1, which cannot create a positive alarm; a missing required history ends the supported prefix. An alarm already raised is not withdrawn by a later failure. Numerical failure handling is part of the calibrated map, not permission to remove difficult paths from the denominator.

A.3 Daily implementation

The daily prior fit uses $\mu \in [-0.1, 0.1]$ per session, $w \in [10^{-8}, 0.25]$ in daily squared-log-return units, $g \in [0, 2]$, and $\nu \in \{6, 10\}$. For each ν, three deterministic starts use $g = 0, 1, 2$. The optimizer scales μ, g linearly and $\log w$ linearly to unit intervals. It uses L-BFGS-B with fixed tolerances, and accepts a solution only after re-evaluating a finite objective with positive variances. Failed attempts are retained. The selected fit is shared by every subsequent exponent.

The market proxy in (15) uses raw squared returns, not centered returns, intraday integrated variance, or the target day's SPY return. No forward filling is allowed. All 252 prior stock/scale rows are required. The real-data reference predicate uses the last 126 training returns: stock annualized root-mean-square volatility at most 0.30 and drawdown at most 0.15, with corresponding SPY limits 0.20 and 0.10. Reference scored suffixes must finish strictly before the target launch. The selected reference order is a fixed hash order, with at most one launch per symbol per quarter.

The batch score uses (13); the monitor uses (14) and the same monotone bounded transformation on positive log evidence. Each is calibrated separately from its endpoint or its full-path maximum. With insufficient references for the rank, the threshold is infinite. The numerical domains make the procedure reproducible; they are not asserted to cover every true market parameter.

A.4 Intraday implementation

The intraday prior fit uses $\mu \in [-0.01, 0.01]$ per five-minute return, $h_{\text{init}} \in [10^{-12}, 0.04]$, and $\nu \in \{6, 10\}$. The clock requires at least 200 prior sessions with at least 70 finite bars and sufficient observations in each scheduled slot; a declared whole-clock SPY fallback is used where allowed by the protocol. A full session contributes 77 adjacent returns from 78 bar closes. Overnight movement and the return into the first bar are excluded.

Let J_d be the supported scheduled returns of a completed session. The common background state updates as

$$h_{d+1} = \lambda h_d + (1 - \lambda)\frac{1}{|J_d|}\sum_{j \in J_d}\frac{(R_{d,j} - \mu)^2}{c_j}, \qquad \lambda = 2^{-1/20}.$$

The clock and the update coefficient are frozen; the state advances only after the session. The variance at each scored return is (16). The mixture is accumulated over all supported returns, even though the price anchor resets each session. Early closes have planned-closed slots rather than fictitious zero returns.

The record is a retrospective replay of split-adjusted consolidated-tape closes with assumed availability one minute after the bar. Bar closes can be sub-cent even in raw data, so a cent-rounding likelihood does not automatically describe them. The numerical method here does not use the lab's certified rounded-price filter.

A.5 Comparators and differences between experiments

GSADF scans expanding and rolling subsamples for right-tailed Dickey–Fuller evidence [21]. The lab reproduced the published critical values and size tables within simulation tolerance. The published Table 4 power comparison was not reproduced: eight of ten cells were outside tolerance, with incompletely specified simulation details affecting that comparison. Later monitoring comparisons evaluate running statistics using prior-inclusive daily history. The level monitors learn nuisance from that history but begin accumulating their scored evidence at launch. Equal launch dates therefore do not imply equal evidence histories.

The JKP implementation follows the volatility-tail method [13]; several operational choices, including bandwidth scaling and the decision that a fitted function is bounded, are the lab's readings where the paper is not fully explicit. The original licensed tick-data figures were not reproduced. Adverse simulation and daily-data results refer to this implementation and sampling design. They should not be described as a proof that every tail-based method fails under its own assumptions.

The panel explosive-root method [18] and the intraday-signal dating method [20] add further comparisons in the research record. Their targets and data requirements differ from the daily level statistic. We do not combine all their rejection frequencies into a single cross-method size ranking.

B Selected derivations

B.1 The endpoint calculation

For $b_\pi(x) = \pi\sigma(x)/x$, the original diffusion has scale density $\rho = 1$ and scale $s(x) = x - c$. In the martingale criterion, infinity is bad because $s(\infty) = \infty$. At zero, the goodness integrand is

$$\frac{(s(x) - s(0))b_\pi(x)^2}{\rho(x)\sigma(x)^2} = \frac{\pi^2}{x},$$

which is not integrable. Both endpoints are therefore bad for positive exposure.

The auxiliary drift is $\pi\sigma^2(x)/x$, its scale density is $\tilde{\rho}(x) = (x/c)^{-2\pi}$, and

$$\tilde{s}(x) = \begin{cases} c^{2\pi}\dfrac{x^{1-2\pi} - c^{1-2\pi}}{1-2\pi}, & \pi \neq 1/2, \\ c\log(x/c), & \pi = 1/2. \end{cases}$$

For $\pi > 1/2$, the scale has finite upper limit and

$$\frac{\tilde{s}(\infty) - \tilde{s}(x)}{\tilde{\rho}(x)\sigma^2(x)} = \frac{x}{(2\pi - 1)\sigma^2(x)}.$$

For $0 < \pi < 1/2$, it has finite lower limit and the corresponding lower integrand is $x/((1 - 2\pi)\sigma^2(x))$. At one-half both scale endpoints are infinite in their respective directions. Applying the cited endpoint criterion yields Theorem 1. The integrability requirement on $b_\pi^2/\sigma^2 = \pi^2/x^2$ holds on every compact subinterval of $(0, \infty)$.

B.2 From tail completion to testing

Let S be a strict alternative under $P = Q$ and choose $n > s_0$. Keep its coefficient below n and complete it linearly above n. The completed upper integral is infinite, giving a martingale null. The original and completed laws agree until first hitting n by stopped uniqueness in law. A coupling of the stopped observations and their continuations therefore bounds, for a test $0 \leq \varphi \leq 1$,

$$|\mathbb{E}_{P_n}\varphi - \mathbb{E}_P\varphi| \leq P\left(\sup_{t \leq T} S_t \geq n\right) \longrightarrow 0.$$

For finitely many paths, use the probability that any path reaches n. The finite-horizon maximum is finite almost surely, so the bound vanishes. Taking the supremum over null rejection probabilities gives the theorem. Allowing equivalent physical measures requires the additional change-of-measure construction in claim 006; the $P = Q$ subfamily already proves the central obstruction.

B.3 An honest likelihood-ratio guarantee

Condition on a training record. Suppose q is a normalized forecast density for evaluation observations and the true conditional density is p_{θ_0} in the admitted null family. Then

$$E = \frac{q(Y)}{\sup_{\theta \in \Theta_0} p_\theta(Y)} \leq \frac{q(Y)}{p_{\theta_0}(Y)}, \qquad \mathbb{E}_{\theta_0}[E \mid \text{training}] \leq 1.$$

Markov's inequality gives a fixed-record rejection bound at $E \geq 1/\alpha$. The reasoning needs a density on the actual observation space. For rounded records this means cell probabilities or a filtered cell likelihood,

not evaluating a continuous density at the reported tick. Observation noise requires its conditional law or an honest envelope as well.

For a sequential coherent model, predictable normalized forecasts produce a likelihood-ratio martingale relative to each true null law. A score dominated at every time by that martingale inherits its crossing bound. The supremum in the denominator must cover the true whole-record law, including the conditioning convention. A nuisance estimate frozen from training does not generally equal the true nuisance, so the real-data monitors need their empirical calibration instead of an automatic appeal to this argument.

If $E_1, \dots, E_m$ are genuine e-values, e-BH uses the largest k for which the kth largest value is at least $m/(qk)$ and rejects the top k. Its expected false-discovery proportion is at most q under arbitrary dependence [26]. This does not guarantee that a realized list has false-discovery proportion below q, nor justify applying the same threshold to scores whose null expectations are unknown.

B.4 The rank calibration bound

Under exchangeability, the target's rank among $m+1$ records is uniform when ties are randomized; strict exceedance of the reference order statistic can only reduce rejection when ties are retained. Thus the probability of exceeding the kth ordered reference is at most $(m+1-k)/(m+1) \le \alpha$. Apply the argument to a whole path's maximum to bound a crossing within its declared horizon. The proof does not establish exchangeability after selecting calm reference records. That is an external property of the application.

C Research provenance and verification scope

This manuscript is an editorial synthesis of the repository's existing claims and experiments. It neither runs new experiments nor promotes a numerical finding to a theorem. The original research record remains in `paper/PAPER.md`, the claim files, the experiment registrations, and their code and summaries. The source version used for generated tables and figures is recorded in `manuscript/generated/provenance.json`, together with SHA-256 hashes of the input result files and bibliography registry.

C.1 Principal mathematical sources within the project

The following map uses stable claim IDs; theorem numbers in this manuscript are new and do not follow the original paper's numbering. Each claim file states its cited inputs, proof, review, and formal counterpart.

Claims	Material used here
001	Two-endpoint exposure classification and absorbed-Brownian counterexample.
002, 003, 006	Compatible tails, broad testing obstruction, pricing-measure ambiguity, and the local-volatility specialization.
008, 012, 013, 015	Visited-range identity, compact estimation, declared power tails, and shared-tail occupation.
004, 005, 007, 009	Holding-period accounting, finite-grid inheritance, refinement, and convergence when the limit retains its mean.
011, 018, 020	Inverse and leveraged-long liquidation, sampled monitoring bounds.
014, 021	Option residual intervals and incomplete-call information.
010, 016, 019, 044	Basket additivity, killed weights, conditional recovery, dispersion sets, and synthetic estimation.
017	American-call inequality, distributions, and conditional basket tolerance bounds.
051–058	Exact-catalogue inference, price-law instances, e-BH, continuous Gaussian nuisance, and earlier-window parameter declarations.
023–032, 059–061	Misspecification, rounding, and jump/switch extensions.
033–038, 064–067	Clocks, observation noise, exact filtering, quadrature, and cross-sectional clock uncertainty.
062–063	Pooled likelihood constructions; comparisons are reviewer simulations in the claim files.
039–042, 068–070	Sequential scores, funded restarts, predictable implied volatility, and variance-switch profiles.
045	Empirical rank calibration and its post-selection premises.
071–076	Continuous-record elasticity, variation profiles, small-scale obstruction, infinite horizons, and independent assets.

The numerical statement about exact rounded filtering is supported by the claim-036–038 evaluators and checks, not by the Student-t monitor. The exposure extensions use claims 074–076 and their separate checks. They should not be confused with later experiment numbers that happen to use similar direction labels.

C.2 Empirical source map

Every numbered experiment has a registration file directly in `experiments/` and, when run, code and outputs in the corresponding numbered subdirectory. A "reproduced" label below refers to the lab's recorded review, not a fresh full replication performed for this manuscript.

Experiments	Use in this manuscript and principal limitation
001, 006	Exposure examples; numerical evidence about selected models.
007–009, 012	Measured ETF costs, finite rebalancing, and convergence; expectation convergence is not inferred from a path plot alone.
013	Sampled visited-integral estimator; not the same object as the exact continuous-path identity.
017–018	Liquidated funds and clipped monitoring; numerical defect estimates have sampling and monitoring error.
002–005, 015	Comparator implementations and simulation responses; different null targets, and original licensed data unavailable for some methods.
019, 022–028	Exact catalogues, nuisance flexibility, regimes, clocks, and rounding; guarantees stay within the stated observation laws.
021, 030–033	Historical daily, collapsed-name, and hourly comparisons; retrospective labels and incomplete free-data coverage.
029, 034–038	Common clocks, nuisance resets, wider forecasts, and restarts; no general power guarantee follows.
010	Free-data survivorship: no daily series for 57 of 506 January 2020 members, 40 of 505 December 2021 members, and 27 of 503 January 2023 members.
032–033, 035	Earlier hourly likelihood scores: 0 of 21 press-labelled records versus GSADF's 18 of 21; the widened-scale GameStop flag was jump-sensitive.
041	Jump audit: 156 of 185 earlier flags survive filtering; rising within-window volatility is the stronger daily distinction.
039–040, 045	Broad universe and GSADF calibration; current constituents, overlapping windows, selected reference records.
043, 047	Implied-volatility scale; real-data silence does not verify the scale band or identify absence of a bubble.
046	Conditional basket recovery in synthetic terminal samples; fixed-bin bias and supplied smoothness restrict the practical result.
049–051	Diagnosis, repaired-score simulation, and daily replay; research redesign and calibration transport remain relevant.
052–053	Intraday description and estimator validation; coefficient sign is not specific to a local-volatility mechanism.
054	Intraday monitor replay; 499 references, 449 run-up launches, 8,600 broad-draw launches; prior-inclusive GSADF and no untouched holdout.
055	Daily validation; 499 reference and 500 evaluation paths per main null cell; partial validation and large calibration-transport effects.

Experiments	Use in this manuscript and principal limitation
056	Daily replay; 153,716 reported universe launches, including 3,197 without finite calibration; one market proxy and current constituents.

Selected numerical tables are generated directly from the result files for experiments 040, 054, 055, and 056. The calibration-transport and daily-cohort figures are vector plots of those same outputs. The liquidation figure is a schematic. Other numerical summaries in prose are transcribed from the registered reports and the original paper; their cited experiment identifies the relevant design and scope. The build records input hashes and checks the typeset document, but does not refit models, regenerate market data, or rerun the simulations.

C.3 What Lean verifies

The project uses Lean 4 with a pinned Mathlib. Its stochastic inputs, including diffusion existence, martingale criteria, bounded stopping, and occupation identities, enter as explicit hypotheses where the library does not supply the relevant theory. Selected deterministic analysis and conditional implications are proved from those inputs.

For the exposure classification, the endpoint analysis is formalized without stochastic assumptions beyond its explicit deterministic setting. The wealth classification is a conditional theorem from the stated martingale criterion and representation. Some upstream structures have only vacuous instances because the pinned library does not construct the needed Brownian model; others have genuine but degenerate instances. They do not establish a nontrivial diffusion instance or add evidence that the supplied stochastic premises hold.

The unconditional tail-completion algebra, several finite-dimensional interval calculations, and portions of the estimator mathematics have stronger formal coverage than the stochastic interpretations. Conversely, some prose corollaries—such as replacing a finite lower-endpoint integral by the event of reaching zero—use cited stochastic results beyond their formal counterparts. The exact scope is documented in `paper/PROOFMAP.md`, the original paper's Appendix A, and `board/FIDELITY.md`.

Formal verification increases confidence in the statements actually encoded. It does not establish their novelty, the fidelity of every translation from a source, a model of actual traded contracts, or the validity of market-data assumptions. No new Lean theorem is claimed for this manuscript's empirical constructions.

C.4 Literature positioning

The mathematical results apply established martingale criteria, diffusion analysis, share-measure constructions, and stochastic-integral facts [6, 15, 16, 19, 23]. Likelihood envelopes, e-BH, rank calibration, and the option valuation conventions likewise have established antecedents. The contribution lies in the exposure classification, the precise identification results, the signal designs, and their measured behavior. No separate novelty claim is made for the finite-trading results.

Section 2 supplies the substantive prior-art comparison. For this manuscript, the author versions of Cox–Hobson, Herdegen–Herrmann, and Herdegen–Schweizer and the Federal Reserve working paper by Gürkaynak were retrieved and checked for the background statements cited there. Copies and edition metadata are retained in `manuscript/sources/`. In particular, the Herdegen–Herrmann citation uses the 2017 arXiv version read here, rather than the earlier working-paper year in the research registry. This

comparison establishes the context of the contributions; it is not an exhaustive theorem-by-theorem novelty audit. Other references retain the repository's version years where a working paper was used.

References

[1] Marco Avellaneda and Stanley Zhang. Path-dependence of leveraged ETF returns. SIAM J. Financial Math. 1 (2010); NYU working copy of May 19, 2009., 2009. URL https://doi.org/10.2139/ssrn.1404708.

[2] Federico M. Bandi and Peter C. B. Phillips. Fully Nonparametric Estimation of Scalar Diffusion Models. Econometrica 71(1):241-283., 2003. URL https://doi.org/10.1111/1468-0262.00395.

[3] Erhan Bayraktar, Constantinos Kardaras, and Hao Xing. Strict local martingale deflators and valuing American call-type options. Finance and Stochastics 16(2):275-291; arXiv:0908.1082., 2012. URL https://doi.org/10.1007/s00780-011-0155-y.

[4] Peter Carr, Travis Fisher, and Johannes Ruf. On the Hedging of Options On Exploding Exchange Rates. arXiv, 2012. URL https://arxiv.org/abs/1202.6188v3.

[5] Alexander M. G. Cox and David G. Hobson. Local martingales, bubbles and option prices. Finance and Stochastics 9:477–492; author manuscript, 2005. URL https://warwick.ac.uk/fac/sci/statistics/staff/academic-research/hobson/publications/bubblesfs.pdf.

[6] Alison Etheridge. Continuous martingales and stochastic calculus. University of Oxford lecture notes, March 11, 2018., 2018. URL https://www.maths.ox.ac.uk/system/files/attachments/b8.2-NewNotesAME.pdf.

[7] Danielle Florens-Zmirou. On estimating the diffusion coefficient from discrete observations. J. Appl. Probab. 30(4):790-804., 1993. URL https://doi.org/10.2307/3214513.

[8] Lawrence R. Glosten, Ravi Jagannathan, and David E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. The Journal of Finance 48(5):1779-1801., 1993. URL https://faculty.washington.edu/ezivot/econ589/GJRJOF1993.pdf.

[9] Robin Greenwood, Andrei Shleifer, and Yang You. Bubbles for Fama, 2017. URL https://doi.org/10.3386/w23191.

[10] Refet S. Gürkaynak. Econometric Tests of Asset Price Bubbles: Taking Stock. Finance and Economics Discussion Series 2005-04, Federal Reserve Board, 2005. URL https://www.federalreserve.gov/pubs/feds/2005/200504/200504pap.pdf.

[11] Martin Herdegen and Sebastian Herrmann. Strict Local Martingales and Optimal Investment in a Black–Scholes Model with a Bubble. arXiv:1711.06679v1, 17 November 2017, 2017. URL https://arxiv.org/pdf/1711.06679v1.

[12] Martin Herdegen and Martin Schweizer. Strong bubbles and strict local martingales. International Journal of Theoretical and Applied Finance 19, 1650022; author version of 28 May 2016, 2016. URL https://people.math.ethz.ch/~mschweiz/Files/bubbles-final-hp.pdf.

[13] Robert Jarrow, Younes Kchia, and Philip Protter. How to Detect an Asset Bubble. SIAM J. Financial Math. 2:839-865; Johnson School Research Paper No. 28-2010 (SSRN 1621728), human-supplied., 2011. URL https://doi.org/10.1137/10079673x.

[14] Robert A. Jarrow, Philip Protter, and Kazuhiro Shimbo. Asset Price Bubbles in Complete Markets. In Advances in Mathematical Finance, Birkhäuser (author-hosted copy, ms.mcmaster.ca)., 2007. URL https://doi.org/10.1007/978-0-8176-4545-8_7.

[15] Ioannis Karatzas and Johannes Ruf. Distribution of the time to explosion for one-dimensional diffusions. Probability Theory and Related Fields, 2015. URL https://doi.org/10.1007/s00440-015-0625-9.

[16] Constantinos Kardaras, Dörte Kreher, and Ashkan Nikeghbali. Strict local martingales and bubbles. The Annals of Applied Probability, 2015. URL https://doi.org/10.1214/14-aap1037.

[17] Eiji Kurozumi, Anton Skrobotov, and Alexey Tsarev. Time-Transformed Test for the Explosive Bubbles under Non-stationary Volatility, 2021. URL https://doi.org/10.2139/ssrn.3755872.

[18] Yanbo Liu, Peter C. B. Phillips, and Jun Yu. A Panel Clustering Approach to Analyzing Bubble Behavior. Cowles Foundation Paper 1862 (Int. Econ. Review)., 2022.

[19] Aleksandar Mijatović and Mikhail Urusov. On the martingale property of certain local martingales. Probability Theory and Related Fields, 2010. URL https://doi.org/10.1007/s00440-010-0314-7.

[20] Yoshiki Obayashi, Philip Protter, and Shihao Yang. The Lifetime of a Financial Bubble. SSRN 2618486, version of March 17, 2016 (Math. Financ. Econ. 2017); human-supplied., 2016. URL https://doi.org/10.2139/ssrn.2618486.

[21] Peter C. B. Phillips, Shu-Ping Shi, and Jun Yu. Testing for Multiple Bubbles. Cowles Foundation Discussion Paper 1843 (working-paper version of the Int. Econ. Review 2015 paper 'Historical episodes of exuberance and collapse in the S and P 500')., 2012.

[22] Peter C. B. Phillips, Shu-Ping Shi, and Jun Yu. Testing for Multiple Bubbles: Limit Theory of Real Time Detectors. Cowles Foundation Discussion Paper 1915 (working-paper version of the Int. Econ. Review 2015 paper)., 2013. URL https://doi.org/10.2139/ssrn.2327633.

[23] Johannes Ruf. The martingale property in the context of stochastic differential equations. Electronic Communications in Probability, 2015. URL https://doi.org/10.1214/ecp.v20-3449.

[24] Carlos A. Sin. Complications with Stochastic Volatility Models. Cornell University, Operations Research and Industrial Engineering Technical Report 1139, December 1995., 1995. URL https://hdl.handle.net/1813/9021.

[25] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. NeurIPS 2019 (arXiv 1904.06019)., 2019. URL https://arxiv.org/abs/1904.06019.

[26] Ruodu Wang and Aaditya Ramdas. False discovery rate control with e-values. arXiv, 2020. URL https://arxiv.org/abs/2009.02824v5.

[27] Larry Wasserman, Aaditya Ramdas, and Sivaraman Balakrishnan. Universal Inference. arXiv, 2019. URL https://arxiv.org/abs/1912.11436v4.

[28] Michel Émery. Équations différentielles stochastiques lipschitziennes : étude de la stabilité. Séminaire de probabilités (Strasbourg) 13, 281–293. Numdam open archive., 1979. URL https://www.numdam.org/article/SPS_1979__13__281_0.pdf.